Engineering Applications of Artificial Intelligence 181 (2026) 115649



Contents lists available at ScienceDirect

Engineering Applications of Artificial Intelligence

journal homepage: www.elsevier.com/locate/engappai





Deep reinforcement learning-based decision support system for optimal maintenance planning in bridge digital twins

Vahid Mousavi [a,*], Maria Rashidi [a,b,**], Shayan Ghazimoghadam [c], Masoud Mohammadi [a], Bijan Samali [a]

[a] Centre for Infrastructure Engineering (CIE), Western Sydney University, Penrith, NSW, 2751, Australia
[b] Urban Transformations Research Centre (UTRC), Western Sydney University, Parramatta, NSW, 2150, Australia
[c] Department of Civil Engineering, Sha.C, Islamic Azad University, Shahrood, Iran

ARTICLE INFO

Keywords:
Bridge management system
Bridge maintenance planning
Decision support system
Reinforcement learning
Multi-objective reinforcement learning
Markov decision process
Digital twins

ABSTRACT

Bridge infrastructures are inevitably exposed to a wide range of deterioration mechanisms and external damages over their service life, which can gradually impair structural integrity, compromise functionality, and increase safety risks. These challenges, coupled with the growing complexity of infrastructure systems and limited financial resources, highlight the need for implementing intelligent and proactive asset management solutions. To address these needs, this paper presents a Reinforcement Learning-based Decision Support System (RL-DSS) framework for bridge maintenance optimization, leveraging a Multi-Objective Deep Centralized Multi-Agent Actor-Critic (MO-DCMAC) algorithm that is designed for application within bridge Digital Twin environments. The proposed RL-DSS implements a three-objective MO-DCMAC framework that models maintenance cost, structural failure risk, and serviceability as separate objectives within a vector-valued reward formulation, while employing a fixed linear utility function for policy optimization. The proposed framework was evaluated through three real-world bridge case studies, each representing different service conditions, to demonstrate its capability in providing long-term maintenance planning. Its performance was benchmarked against multiple established rule-based policies commonly used in practice. The results revealed that the MO-DCMAC agent achieved the highest overall utility score, demonstrating superiority by generating more refined and practical maintenance policies. By actively penalizing high-frequency site visits and strategically balancing intervention costs with risk mitigation, the RL agent prioritizes fewer but more impactful repair actions. The proposed RL-DSS reflects an adaptive, informed, and cost-effective decision-making process that captures the complexity of real-world maintenance scenarios, surpassing the limitations of traditional rule-based approaches.

1. Introduction

Sustainable transportation systems are widely recognized as critical foundations for the development of resilient and sustainable societies, and are closely aligned with the principles of smart cities (Moslem et al., 2024; Saha et al., 2024). Achieving these sustainability objectives relies heavily on the implementation of effective infrastructure asset management systems, which enable decision-makers to apply, data-driven, timely and predictive maintenance strategies supported by advanced intelligent control and autonomous management approaches (Görçün et al., 2025; Sood, 2024; Tang et al., 2025). In this context, bridges serve as fundamental components of transportation infrastructure and the broader built environment. However, their structural condition deteriorates over time due to various factors, including environmental exposure, traffic-induced loading, material degradation and aging. Without appropriate monitoring and management, these factors can lead to reduced structural capacity and serviceability, which may necessitate major rehabilitation, repair, or even demolition, resulting in increased costs, prolonged traffic disruptions, and elevated safety risks (Wu et al., 2021). Therefore, considerable attention has been directed

This article is part of a special issue entitled: AI in Control published in Engineering Applications of Artificial Intelligence.

* Corresponding author.
** Corresponding author. Centre for Infrastructure Engineering (CIE), Western Sydney University, Penrith, NSW, 2751, Australia.
E-mail addresses: v.mousavi@westernsydney.edu.au (V. Mousavi), m.rashidi@westernsydney.edu.au (M. Rashidi), Shayan.ghazimoghadam@iau.ir (S. Ghazimoghadam), Masoud.Mohammadi@westernsydney.edu.au (M. Mohammadi), b.samali@westernsydney.edu.au (B. Samali).

https://doi.org/10.1016/j.engappai.2026.115649
Received 3 September 2025; Received in revised form 10 June 2026; Accepted 6 July 2026
Available online 10 July 2026
0952-1976/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

toward evaluating the structural health and implementing proactive maintenance strategies for critical infrastructure assets (Wei et al., 2020).

Over the years, a range of maintenance approaches has been developed to enhance infrastructure longevity and reliability. Proactive maintenance strategies aim to prevent deterioration through timely and planned interventions. One common approach is time-based maintenance, which involves performing scheduled repairs at predetermined intervals, regardless of the structure's actual condition. Although straightforward to implement, time-based methods often lead to inefficiencies by triggering unnecessary maintenance actions (de Jonge et al., 2017). In contrast, condition-based maintenance, which relies on observed or predicted structural states derived from inspection data or analytical models, tends to be more efficient than time-based maintenance due to its reliance on inspection information and tailoring interventions to the asset's actual condition and performance needs (Zonta et al., 2020).

However, the rapid expansion of transportation infrastructure and advances in technology have considerably reshaped the landscape of bridge management. Preventive maintenance, particularly for bridges with different structural forms, could be a challenge for engineers, especially in the absence of a structured, data-informed decision-making frameworks (Brighenti et al., 2024; Rashidi et al., 2016a). This is mainly because strategic bridge management and maintenance extends beyond routine preventive maintenance needs; it requires prioritizing long-term safety, operational functionality, and economic sustainability (Frangopol and Kim, 2022). Given the substantial financial and environmental implications associated with replacing deteriorating bridge structures, extending the service life of existing bridges has become a more cost-effective and sustainable alternative.

In recent decades, decision-making in bridge asset management has become progressively more complex, largely due to structural complexities and the integration of advanced digital and communication technologies. These advancements have expanded the range of potential management solutions, complicating the identification of optimal strategies for asset managers. As a result, a number of specialized Bridge Management Systems (BMSs), such as PONTIS (Thompson et al., 1998), DANBRO(Andersen, 1990) and BRIDGIT (Hawk and Small, 1998), have been developed to support systematic maintenance planning and asset preservation. These systems integrate data inputs, regulatory standards, and specialized software tools to facilitate tasks such as maintenance planning, prioritization of interventions, and lifecycle cost optimization (Brighenti et al., 2024; Rashidi, 2013).

Emerging technologies including Bridge Information Modelling (BrIM), Digital Twins (DTs) and Deep Learning (DL) are further advancing the BMS capabilities. By enabling the digital representation of assets and integrating real-time monitoring data with predictive models, these technologies allow for reliable assessment of structural health and more informed decisions regarding the timing and scope of maintenance activities (Mousavi et al., 2025a, 2025b). Consequently, transportation authorities around the world are increasingly adopting these systems to address the growing complexity of infrastructure management and improve the efficiency and effectiveness of bridge maintenance practices (Shahrivar et al., 2025).

Despite significant advancements in development of bridge DTs, application of Decision Support Systems (DSS) into bridge management systems has been rarely considered in bridge DTs (Mousavi et al., 2024). Whereas they can provide powerful capabilities for enhancing maintenance efficiency, improving decision-making quality, and reducing the time and cost associated with planning interventions. Therefore, when effectively integrated, they could serve as the core of a DT, synthesizing the data gleaned from monitoring, predictive modelling, simulation, and lifecycle analysis into actionable insights. Nonetheless, the current implementations of bridge DTs often lack robust bidirectional strategies capable of translating analytical outputs into actionable maintenance interventions.

Reinforcement learning (RL) has emerged as a promising methodology for enhancing prescriptive maintenance strategies (Saifullah et al., 2024). In the context of infrastructure management, RL formulates the maintenance planning problem as a sequential decision-making problem, where each decision influences both immediate outcomes and future system performance. RL algorithms learn optimal strategies through iterative interactions with the environment, guided by reward signals that quantify the effectiveness of decisions. The primary goal is to maximize cumulative rewards over time, reflecting long-term performance and cost efficiency (Sutton and Barto, 1998). The ability of RL to accommodate complex infrastructure portfolios, support long-term optimization over extended planning horizons, and operate effectively under uncertainty highlights its potential to transform infrastructure maintenance and asset management practices. While RL offers a theoretically sound framework for intelligent maintenance planning under uncertainty, existing methodologies are often limited by the absence of effective algorithms for deriving optimal policies. A critical review of the literature reveals persistent limitations in achieving key operational objectives, as most existing RL-based approaches predominantly focus on single objective optimization (Heng et al., 2024). In practice, however, maintenance optimization problems inherently involve multiple, and often conflicting, objectives that must be considered simultaneously. These limitations emphasize the necessity for dedicated research efforts that bridge the gap between theoretical RL constructs and their practical implementation in real-world asset management systems.

This study introduces an advanced RL-based Decision Support System (RL-DSS) for bridge maintenance optimization using Multi-Objective Deep Centralized Multi-Agent Actor-Critic (MO-DCMAC). The framework employs MO-DCMAC to simultaneously handle uncertainties and incorporating multiple, distinct penalty signals through a weighted linear utility formulation. In the proposed framework, the environment produces a vector-valued reward corresponding to maintenance cost, structural risk, and serviceability, while the weighted utility function is applied during policy optimization to derive the scalar decision signal used for policy improvement. This approach enables intelligent agents to learn effective maintenance policies while considering component interactions, and multiple criteria, including material properties, bridge age, traffic loading, environmental conditions, and long-term structural health. A multi-criteria decision-making process is integrated into the framework to balance cost, risk, and performance objectives. The main contributions and novelties of this study are as follows:

1. Development of a multi-objective RL-based Decision Support System (RL-DSS) for bridge maintenance, designed to operate as a decision-making module within DT-based bridge management environments.
2. The bridge maintenance problem is formulated as a Multi-Objective Markov Decision Process (MOMDP), with maintenance cost, structural risk, and serviceability represented as separate reward components. A fixed linear utility function is used for policy optimization, while the vector-valued representation is retained to preserve detailed per-objective analysis.
3. The MO-DCMAC framework is adapted from two to three objectives as an implementation-level contribution. A vectorized joint distributional critic and a multi-dimensional Bellman projection are developed to ensure computationally efficient learning in the three-objective setting.
4. Empirical validation across multiple real-world bridge case studies and comparison of the proposed RL-DSS performance with the multiple established rule-based policies.

A series of case study evaluations were conducted using real-world bridge case studies to assess the effectiveness of the proposed framework in long-term maintenance planning. The results demonstrate that RL-DSS, powered by the MO-DCMAC algorithm, can effectively balance

multiple objectives such as cost efficiency, risk mitigation, and structural performance. These findings highlight the potential of the proposed framework to enhance decision-making in bridge management systems and serve as a foundation for future advancements in intelligent infrastructure maintenance.

The remainder of this paper is organized as follows: Section 2 reviews related works and outlines the motivation of the proposed approach. Section 3 provides a brief overview on the RL, multi-objective RL and MO-DCMAC methods. Section 4 introduces the proposed RL-DSS solution, while Section 5 details the deep RL methodology and the MO-DCMAC model. Section 6 reports real bridge case study results. Section 7 compares the results against established maintenance policies, Section 8 presents a parameter analysis, while Section 9 discusses potential limitations. Finally, Section 10 concludes the paper with final remarks and future research directions.

2. Related works and motivations

In bridge maintenance optimization, decision-making can be structured at either the network level or the individual asset level. Network-level strategies optimize maintenance across multiple assets, typically applying uniform interventions to all structures (Jaafaru and Agbelie, 2022; Lei et al., 2022; Xu and Guo, 2022). SHM data and sustainability criteria could also be combined into a single utility-based reward to enable efficient life-cycle maintenance policies across multiple bridges at network level (Lei et al., 2023, 2025). In contrast, bridge-level optimization, which is the focus of this study, targets a single structure and concentrates on the strategic maintenance of its internal components. Although the scale is smaller, bridge-level optimization introduces its own complexities, particularly the compounded risks that arise when multiple critical subsystems degrade simultaneously.

A review of existing literature reveals that many bridge-level maintenance strategies have been proposed. These methods are predominantly grounded in classical optimization approaches, rule-based systems, or supervised machine learning techniques. For example, Wang et al. (2024) proposed a data-driven maintenance optimization framework for bridge infrastructure that integrates reliability-based deterioration, intervention, and financial models. Similarly, Ghafoori et al. (2024) introduced a system that integrates ML techniques and binary linear programming to predict the condition of concrete bridge elements and optimize maintenance interventions. Using historical data, the system models deterioration and generates long-term plans under budget constraints. Other frameworks, such as the ontology-driven smart contract system for bridge preservation decision-making proposed by He et al. (2025), focus on the gap in formalizing and applying implicit rules derived from expert judgment. Their method integrates domain ontology with smart contracts to enable transparent rule execution and decentralized data sharing. While these approaches offer valuable contributions, they are limited in their capacity to adapt dynamically to evolving bridge conditions and uncertain operational environments.

In contrast, RL methods have emerged as promising tools for adaptive and intelligent infrastructure maintenance planning. These techniques formulate maintenance as a sequential decision-making process, enabling agents to learn optimal policies through environmental interaction. For instance, Wei et al. (2020) developed a DRL-based framework using the Deep Q-Network (DQN) algorithm to optimize bridge maintenance decisions. The model considered five system states and four possible actions per component, with rewards based on maintenance costs and structural risks. In another research, Xin et al. (2023) proposed a dynamic programming-based intelligent agent to optimize component level bridge maintenance decisions using regular inspection data. A two-parameter Weibull distribution predicts the duration of bridge elements in various states, enhancing cost estimation and tailoring decisions to specific defects.

However, a critical observation from the literature shows that Multi-Objective Reinforcement Learning (MORL) methods tailored for bridge maintenance planning remain relatively scarce. While most existing RL applications in bridge management focus on optimizing single-objective reward structures, real-world maintenance scenarios inherently involve multiple, often conflicting objectives. Only a few studies have explicitly formulated the problem within a MORL framework. For example, Du and Ghavidel (2022) proposed a parameterized RL framework that generates a unified maintenance policy for diverse highway bridges. The framework simultaneously optimizes multiple objectives by integrating a time- and policy-dependent life-cycle risk assessment, balancing maintenance cost and structural risk over time. In another study, van Remmerden et al. (2025) presented a Multi-Objective Deep Reinforcement Learning (MODRL) framework for infrastructure maintenance that directly optimizes policies under nonlinear utility functions. The method is tested using threshold-based and FMECA-based utility functions across multiple maintenance environments, including historical quay walls in Amsterdam. Results show MO-DCMAC significantly outperforms conventional heuristic rule-based policies, demonstrating its effectiveness in balancing cost and structural safety.

Beyond the MORL, limited research integrates intelligent DSS methods within DTs frameworks for predictive and data-driven bridge management. For example, Heng et al. (2024) proposed a DT-based maintenance framework for ageing steel bridge hangers, integrating monitoring data, inspections, and predictive modelling. Their method integrates Partially Observable Markov Decision Process (POMDP), Asynchronous Advantage Actor Critic, and Bayesian Dynamic Linear Model algorithms to guide proactive interventions such as early repairs and strategic replacements, thereby improving maintenance efficiency through coordinated, condition-based, and cost-effective decision-making. Table 1 presents the comparison of solutions reported in recent literature for bridge maintenance optimization problem. The gaps in these methods include the limited application of MORL tailored for bridge infrastructure maintenance, the lack of application of RL-based approaches within DT technologies, and the insufficient ability of existing models to dynamically adapt to real-time structural conditions and evolving operational demands. This study aims to address these key challenges through the development of a novel RL-DSS framework that can be seamlessly integrated with bridge digital twins to provide a robust and adaptive decision-making tool.

3. Theoretical background

This section provides a brief overview of the fundamental concepts and background related to the DL algorithms employed in the proposed RL-DSS. It begins with an introduction to RL, followed by a brief explanation of MORL and DCMAC.

3.1. Brief overview on reinforcement learning

RL involves an autonomous agent that learns optimal strategies for achieving predefined objectives through continuous interaction with its environment. RL does not require prior data for training; instead, the agent iteratively explores its environment, evaluates the outcomes of its actions through a reward-based feedback mechanism, and refines its decision-making policy. The primary objective in RL is to maximize the cumulative reward over time, enabling the agent to identify and execute actions that yield the most beneficial long-term outcomes. This problem can be systematically modelled using a Markov Decision Process (MDP) (Sutton and Barto, 1998), which provides a mathematical framework for decision-making under uncertainty and over extended time horizons.

In an MDP, the state transitions and rewards are defined by a tuple (S, A, P, R, γ), where S is the state space, A is the action space, P defines transition probabilities, R is the reward function, and $\gamma \in [0,1]$ is the discount factor that determines the importance of future rewards. The Bellman expectation equation provides a recursive definition for the state-action value function Q under a policy π:

Table 1
Comparison of the recent solutions for bridge maintenance optimization problem.

Ref	Approach	RL Applied?	Multi-Objectivity Integrated?	DT application Capability
Heng et al. (2024)	Combines inspection data with a DT for fatigue and corrosion prediction and maintenance scheduling	×	✓	✓
Xin et al. (2023)	Agent-based sequential decision model to optimize bridge component actions	×	✓	×
He et al. (2025)	Ontology-driven smart contract for rule-based decision automation	×	×	×
Wang et al. (2024)	Improved Electric Fish Optimization for reliability-based maintenance	×	✓	×
van Remmerden et al. (2025)	Multi-objective Deep actor–critic framework (MO-DCMAC) for utility-based maintenance policy learning	✓	✓	×
Ghafoori et al. (2024)	ML model optimizing performance under annual budget constraints	×	✓	×
Du and Ghavidel (2022)	Parameterized DRL to generate a unified maintenance policy for bridge portfolio by integrating a time and life-cycle risk	✓	✓	×

$$Q^{\pi}(s,a) = E(R(s,a) + \gamma Q^{\pi}(s',a')|s,a) \tag{1}$$

Using MDP, at each time, the agent selects an action and receives a corresponding reward. The reward function provides feedback on action quality. The objective of the RL agent is to maximize the expected cumulative reward over time, with future rewards discounted by a factor $\gamma \in [0,1]$ to reflect their relative importance to the present decision-making context.

A widely used model-free RL method is Q-learning (Watkins and Dayan, 1992), where the Q-function is updated iteratively as:

$$Q(s,a) \leftarrow Q(s,a) + \alpha\left[r + \gamma \max_{a'}(s',a') - Q(s,a)\right] \tag{2}$$

The optimal policy maximizes the expected return from each state-action pair:

$$Q^{*}(s,a) = arg\,max\,E(R_t|S_t = s, a_t = a, \pi) = arg\,max\,E\left(\sum_{k=1}^{H}\gamma^k r_k \middle| S_t = s, a_t = a, \pi\right) \tag{3}$$

RL methods are classified as model-based, which uses a known model of the environment to plan actions efficiently but can be sensitive to model inaccuracies, and model-free approaches, which learn optimal behaviour through trial and error without a predefined model.

RL has experienced significant advancements due to the emergence of novel algorithms capable of tackling increasingly complex problems. In this context, complexity is primarily associated with high-dimensional state and action spaces, which often pose limitations for traditional RL methods. DL techniques have effectively mitigated these challenges through Deep Reinforcement Learning (DRL) that leverages deep neural networks to manage and learn from complex, high-dimensional environments.

In DRL, neural networks are used to approximate value functions, such as $Q(s,a;\theta)$ where θ are the parameters of the Q-network. One example is the DQN, where the temporal difference error is minimized by adjusting θ through backpropagation, enabling generalization across large, continuous state spaces.

3.2. Multi-objective reinforcement learning (MORL)

Traditional RL approaches are inherently designed to optimize policies for single-objective tasks. However, many real-world problems involve the simultaneous consideration of multiple objectives, which often differ fundamentally in nature and cannot be readily combined into a single scalar reward. These objectives represent diverse dimensions of system performance that are neither directly comparable nor easily reducible to a unified metric. Consequently, addressing the complex and multi-dimensional nature of infrastructure maintenance demands more advanced and flexible learning frameworks.

To address this, MORL extends the RL framework to accommodate multiple goals. MORL models are formalized through a multi-objective MDP, where the key distinction from a conventional MDP lies in the reward function which yields a vector of rewards across multiple objectives rather than a single scalar.

Unlike conventional RL, assessing policy performance in MORL presents inherent complexity. This arises from the presence of multiple, often conflicting, objectives. For instance, one policy may improve one objective while another improves a different one, making comparisons ambiguous without understanding of the relative importance of objectives. This challenge is particularly evident in bridge maintenance planning, where decision-makers must weigh trade-offs.

In such cases, a key concept is Pareto optimality. A policy is considered Pareto-optimal if no other policy can improve one objective without worsening another. The set of all such non-dominated policies forms the Pareto front, providing a visual and analytical tool for evaluating trade-offs.

To address this, utility-based approaches are commonly employed to aggregate the vector of cumulative rewards into a single scalar utility value using a predefined function. This scalarization enables consistent evaluation and comparison of policies within the multi-objective framework. A common form of scalarization is the linear scalarization method, where:

$$u(R) = W^{T}R = \sum_{i=1}^{n} w_i R_i \tag{4}$$

where w represents the weight vector that reflects the relative importance of each objective.

For effective optimization, the utility function must exhibit strict monotonicity, which ensures that improvements in any objective always increase overall utility, enabling continuous policy optimization and preventing convergence to suboptimal solutions.

One approach for optimizing policies in MORL under a known utility function is the Expected Scalarized Return (ESR) criterion (Hayes et al., 2022a). The ESR criterion evaluates the utility of the cumulative return associated with each trajectory prior to computing the expectation across trajectories. Formally, the ESR criterion seeks to find a policy that maximizes:

$$\pi^{*} = arg\,max\,E\left(u\left(\sum_{k=1}^{H}\gamma^k r_k\right)\middle| S_t = s, \pi\right) \tag{5}$$

In Eq. (5), the utility function u is applied to the total return vector of a single trajectory before the expectation is taken across multiple trajectories. This means that the evaluation reflects the performance of each run individually, rather than averaging rewards first and then computing utility.

Two evaluation criteria are commonly distinguished in MORL. Scalarized Expected Return (SER) first computes the expected vector return and then applies the utility function, while ESR applies the utility function to each trajectory return before taking the expectation. It should be noted that this distinction between ESR and SER becomes relevant only when the utility function is nonlinear. Under a fixed linear utility function, however, the two criteria yield mathematically

equivalent policy optimization objectives, as linearity commutes with the expectation operator (Hayes et al., 2022b). Therefore, both criteria yield identical policy optimization objectives in the present formulation.

3.3. Deep Centralized Multi-Agent Actor-Critic (DCMAC)

DCMAC is a RL framework designed for multi-agent systems that combines centralized training with decentralized execution. This approach addresses the key challenges of multi-agent (MA) RL, particularly the non-stationarity problem that arises when multiple agents learn simultaneously in a shared environment (Lowe et al., 2017). This setup follows the Centralized Training with Decentralized Execution paradigm, allowing agents to learn coordinated strategies during training while acting independently during deployment. DCMAC uses a centralized critic with global information for training, while agents execute local actor policies, enabling coordinated learning alongside independent action.

Considering n control units, each capable of taking individual actions. These units may represent either components or sub-systems. The key assumption in DCMAC is conditional independence of actions, given the system state s. This allows the joint policy $\pi(a|s)$ to be factorized as:

$$\pi(a|s) = \prod_{i=1}^{n} \pi_i(a_i|s) \tag{6}$$

where $\pi_i(a_i|s)$ is the probability of action a_i for the *i-th* control unit, based on the shared system state s.

The actor network in DCMAC is centralized in the sense that it receives the entire system state as input and outputs action distributions for all control units. The policy gradient for updating the actor network is computed as:

$$\nabla_\theta J(\theta) = \mathrm{E}_{s,a\sim\mu}\left[\sum_{i=1}^{n} \nabla_\theta \log\ \pi_i(a_i|s) \cdot \mathrm{A}(s,a)\right] \tag{7}$$

where θ are the parameters of the centralized actor network and $\mathrm{A}(s,a)$ is the advantage function, representing how much better an action is compared to the expected value.

The advantage function is defined as $A(s,a) = Q(s,a) - V(s)$, which reduces variance during learning and stabilizes updates.

The critic is trained to minimize the loss:

$$L(\phi) = E\left[\left(\mathrm{Q}_\phi(s,a) - y\right)^2\right] \tag{8}$$

where $y = r + \gamma Q_{\phi'}(s', a')$, and ϕ represents the parameters of the critic network. $Q_{\phi'}$ is also the target network used for stable updates.

By leveraging centralized critics and decentralized actors, DCMAC facilitates scalable and coordinated learning in complex, MA environments, making it suitable for applications in infrastructure management, robotic teams, and distributed control systems.

4. Proposed RL-DSS framework

The application of deep RL and DSS within digital twinning enhances bridge optimal maintenance planning by enabling informed and predictive decision-making. To fully exploit these advantages, a RL-based DSS reference framework is essential.

The proposed RL-DSS is designed to be integrated into data-driven DT environments, where data from diverse sensors are processed using advanced techniques to provide a real-time representation of the bridge's condition. In this framework, the DT utilizes its internal state-estimation and data assimilation algorithms to process, filter, and fuse real-time structural health monitoring data, thereby generating an updated and reliable system-state representation that is subsequently supplied to the RL-DSS for decision making. This continuously updated bridge state information serves as the input for decision-making within the proposed RL-DSS framework in this study. As the condition of bridge elements changes based on real-time data streams, the decision-making process within the RL-DSS dynamically selects actions according to the updated system states using a pre-trained policy, without online retraining or parameter updating. The primary focus of this research is on the RL-based decision-making process, designed to be integrated into data-driven DTs, while detailed explanations of damage detection strategies within DTs are beyond the scope of this study and can be found in recent related works such as Mousavi et al. (2025b).

The proposed framework, shown in Fig. 1, presents a simulation-based methodology to optimize bridge maintenance strategies using a Multi-Objective Deep Reinforcement Learning (MODRL) framework. This approach addresses the complexity of infrastructure management decisions where multiple conflicting objectives must be balanced simultaneously. The primary objectives include minimizing maintenance costs, managing expected failure consequences, and maintaining structural serviceability (penalizing poor DS). The framework employs a MA architecture where individual agents manage distinct bridge components, enabling distributed decision-making within a unified learning structure. This design captures interdependencies between structural elements while maintaining computational efficiency across complex bridge systems.

As illustrated in Fig. 1, the methodology follows an iterative process beginning with bridge condition initialization, incorporating damage indices, material properties, environmental factors, and component ages. The system operates within a dynamic simulation environment that continuously reflects evolving infrastructure states, accounting for both maintenance interventions and natural deterioration processes.

The decision-making process involves systematic observation collection where agents gather state information representing current bridge conditions. Based on these observations and learned policies, agents select maintenance actions from a predefined set ranging from no action to complete component replacement. Selected actions are applied within the simulation environment, triggering state updates that reflect immediate maintenance effects and ongoing deterioration.

A reward calculation mechanism evaluates outcomes across the three primary objectives. This multi-objective penalty-based system quantifies trade-offs between direct cost expenditure, risk thresholds, and serviceability penalties. The framework incorporates a distributional network that estimates long-term consequences by predicting probability distributions over future returns, providing comprehensive evaluation of policy decisions.

Policy optimization integrates calculated rewards and network evaluations through a utility function that enables multi-objective outcome integration. This approach allows the system to learn maintenance strategies aligned with long-term infrastructure management goals while adapting to the complex, time-dependent nature of structural deterioration.

The iterative framework enables continuous learning and adaptation where agents progressively refine decision-making capabilities through repeated simulation interactions. This methodology specifically addresses long-term infrastructure planning challenges where current decisions have consequences extending over decades of operational life. The details of the proposed framework will be discussed in the next sections.

5. Workflow of multi-objective deep RL

This section presents the detailed methodology for the RL-DSS designed for bridge maintenance optimization. It integrates a MO-DCMAC algorithm to balance cost, risk, and serviceability objectives. The following subsections detail the framework's components and technical implementation.

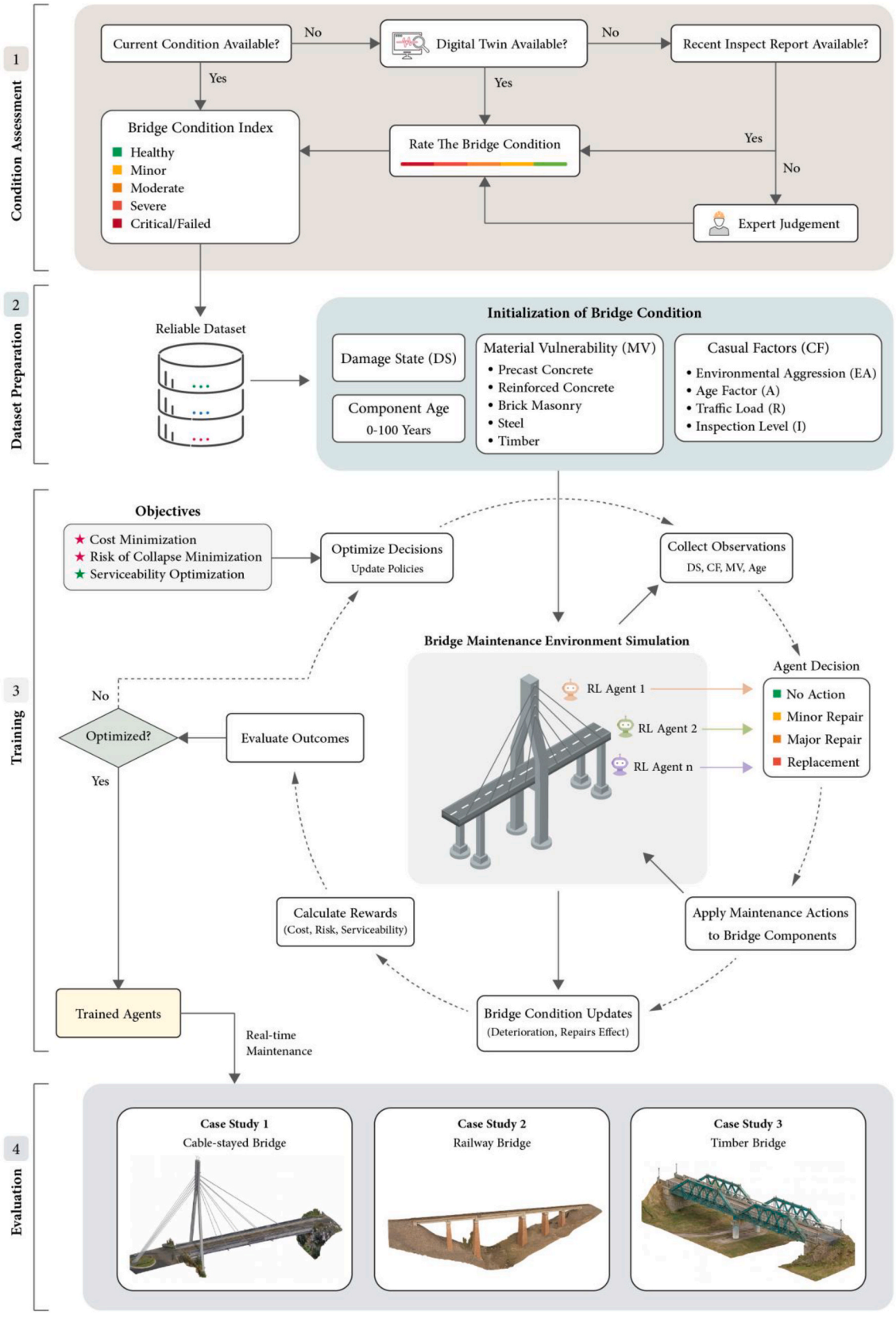


Fig. 1. The overall flowchart of the proposed RL-DSS methodology.

5.1. Bridge condition assessment

As the initial step of the proposed framework, the current bridge condition state is required. The bridge condition or health index is a critical metric for evaluating the structural and functional performance of bridge infrastructure. These general qualitative condition indices are commonly derived from the condition assessment of bridge elements considering the level of service the bridge provides. It plays a pivotal role in identifying deteriorated components and informing the prioritization of maintenance and rehabilitation activities (Chase et al., 2016). The overall condition of each structural element is evaluated using a five-level Damage State (DS) rating scale during the inspection process, as outlined in Table 2.

Traditionally, DS is determined through on-site inspections conducted by qualified inspectors. However, for bridges that are equipped with actively managed DTs, condition assessments can also be derived from these digital models through continuous monitoring and data evaluations. In cases where neither DT systems nor recent inspection reports are available, expert judgment can serve as a valuable alternative for estimating the current condition of the bridge elements. This flexible approach ensures that bridge condition data can still be leveraged to support informed decision-making in asset management, even under data-limited circumstances.

Beyond the assessment of element-level damage states, this study also incorporates two additional technical factors into the evaluation framework, following methodologies established in previous research (Mohammadi et al., 2023; Rashidi et al., 2016a). These include material vulnerability, element importance factor and associated causal factors. Integrating these variables allows for a more comprehensive and reliable bridge maintenance strategy by addressing technical uncertainties and reflecting the differential significance and susceptibility of various components within the overall structural system. The following subsections provide a detailed explanation of each of these factors.

5.1.1. Material vulnerability (MV)

The MV quantifies the susceptibility of different construction materials to environmental degradation and aging. Material type plays a crucial role in determining the durability and long-term performance of bridge components, particularly under harsh service conditions.

While a detailed vulnerability model of bridge elements across all hazard types requires extensive investigations, existing literature and expert assessments indicate that steel components are generally more prone to deterioration than precast reinforced concrete elements under harsh environmental conditions, whereas timber components are particularly vulnerable to biological degradation and moisture-related damage (Rashidi and Gibson, 2011; Valenzuela et al., 2010). Table 3 presents the vulnerability classification of commonly used bridge construction materials.

5.1.2. Causal factors (CF)

Bridge components deteriorate progressively over time and the rate of deterioration is influenced by multiple interacting parameters, including pre-existing conditions, such as original design specifications and construction quality, and post-existing variables related to operational environments, usage patterns, and quality of inspection practices. To systematically capture the impact of these variables on structural degradation, CF criterion is utilized, which is composed of four primary dimensions including environmental exposure, structural age, road type/loading, and inspection quality.

- **Environmental Aggression (EA)**: Environmental factors significantly affect material durability, especially in concrete structures through both chemical and physical processes induced by climatic variables, pollutants, and human activity. These factors can trigger or accelerate deterioration like carbonation, chloride corrosion, and alkali–silica reaction, with early construction defects further facilitating harmful ingress (Rashidi et al., 2016a).
- **Structural Age (A):** The age of a bridge indicates its integrity and fatigue performance. While many bridges have a nominal 50-year design life, some continue to operate up to 100 years, though elements like bearings may need replacement in 25 years. Extended service necessitates re-evaluating safety and durability against original design and current performance standards (Rashidi and Gibson, 2012).
- **Road Type and Loading (R)**: The intensity and nature of traffic loading vary according to a bridge's functional role within the transportation network. In this study, traffic load intensity is approximated using the Average Annual Daily Traffic (AADT) metric. Higher AADT values, characteristic of arterial or highway bridges, are associated with increased structural demand and more rapid wear, whereas bridges on local roads typically experience lower traffic volumes and consequently slower deterioration.
- **Inspection Level (I)**: The quality and consistency of condition assessments are influenced by human factors and inspection practices, including inspector expertise, environment, components accessibility, and adequacy of inspection tools (Ellenberg et al., 2016). In this regard, four distinct levels of bridge inspections have been considered. The initial inspection is conducted after construction. Routine inspections monitor the general condition of the bridge by regular visual evaluations. Detailed inspections use non-destructive testing for potential structural issues, and special inspections are performed following incidents or observed anomalies to support structural analysis for interventions guide (Mohammadi et al., 2023).

Each of the above factors is categorized into four discrete levels, as presented in Table 4.

Table 2
Bridge condition indices.

Index	DS	Description
0	Healthy	No visible deterioration. The element is in as-new condition or defects have no significant structural or functional impact.
1	Minor	Minor defects or early signs of deterioration, with no reduction in the element's functionality.
2	Moderate	Moderate defects and deterioration, resulting in some loss of expected functionality.
3	Severe	Severe defects with significant loss of functionality, or the element is nearing structural failure.
4	Critical	Extensive deterioration or structural damage posing an immediate safety risk; urgent intervention required

Table 3
Material Vulnerability (MV) adopted from Valenzuela et al. (2010).

MV Index	Material Type
1	Precast concrete (non-structural)
2	Reinforced concrete
3	Clay Brick
4	Steel, iron, and alloys
5	Timber

Table 4
Causal Factor (CF) Classification adopted from Rashidi and Gibson (2012).

Level	Environment (EA)	Age (A)	Traffic Load (R)	Inspection Type (I)
1	Mild	0–25 years (recent)	AADT $\leq$150 (minor)	Initial
2	Moderate	25–50 years (new)	150 $<$ AADT $\leq$1000	Routine
3	Harsh	50–75 years (old)	1000 $<$ AADT $\leq$3000	Detailed
4	Very Harsh	75–100 years (very old)	AADT $>$3000 (arterial)	Special

The weighted formulation used to compute the CF is expressed as:

$$CF = 0.12\,EA + 0.41A + 0.11R + 0.36I \tag{9}$$

This equation, provided in our research team study conducted by Rashidi and Gibson (2012); Rashidi et al. (2016a), reflects the relative importance of each contributing factor, with structural age and inspection level criteria receiving the highest weights.

5.1.3. Importance factor (IF)

The IF is used to represent the relative criticality of each bridge element to overall structural safety and service. IF expresses how strongly a component's failure would influence system performance. In practice, IF values are assigned per component based on engineering judgment about structural role, redundancy, and consequence of failure. Within the risk formulation, IF is used as a weight to combine component-level failure probabilities into the probability of system collapse. After each action, updated component failure probabilities are aggregated using normalized IF values, ensuring that failures of more critical elements have a stronger impact on the system-level risk signal.

5.2. Bridge maintenance simulation environment

The bridge maintenance problem is formulated as a MOMDP in this study to provide a structured foundation for implementing sequential decision-making under uncertainties, where multiple conflicting objectives must be optimized over time.

However, the implementation of such frameworks in real-world bridge maintenance encounters data paradox challenge. RL agents require large volumes of diverse data that represent a wide range of system states and the outcomes of various actions in order to learn effective policies. In practice, however, comprehensive and longitudinal real-world data for bridge maintenance is exceptionally scarce and inherently unsuitable for exploratory learning. The reasons for this are threefold:

- **Missing Alternative Outcomes**: Historical records only show what maintenance actions were actually taken, not what would have happened with different choices. Since RL systems learn by comparing different possible actions, this creates a fundamental gap in the available data.
- **Impractical Learning Requirements**: Bridges operate over decades, but RL systems need millions of trial-and-error experiments to learn effectively. Testing different maintenance strategies on real, operational bridges would be both impossible to complete within reasonable timeframes and dangerously risky.
- **Poor Data Quality**: Existing maintenance records are often incomplete, inconsistent, or stored in incompatible formats. This makes them unsuitable for training RL systems, which require reliable and standardized information about system conditions, actions taken, and outcomes achieved.

To overcome these limitations, this study introduces a high-fidelity simulation-based training environment to enable safe, repeatable, and scalable learning for long-term bridge maintenance planning. This digital environment enables the RL agent to interact with a virtual representation of a bridge over countless simulated life cycles, allowing it to develop and refine policies without the risks and constraints associated with real-world testing. The validity of this simulation-based approach hinges on the quality of its underlying models and parameters. The environment for this study is not built on arbitrary rules but is parameterized based on established engineering principles, empirical models, and data reported in existing literature.

Within this simulated framework, the MOMDP is implemented by explicitly defining its fundamental components including the state space, action space, stochastic transition functions and reward structure. Therefore, MOMDP can be defined by a tuple (S,A,P,R,γ), where S is a set of states representing the condition of the bridge. A is set of actions corresponding to maintenance interventions. P is a transition function defining the probability of moving from one state to another given an action. R is a multi-objective reward function that provides feedback on the action's outcome. γ is discount factor for future rewards. In following subsections, the detailed definitions of each component in the proposed framework implementation are presented.

5.2.1. State representation

The primary state variables for each bridge component include a discrete DS, as described in Section 5.1 and Table 2, which quantifies the level of deterioration from *Healthy* to *Critical*, and the component's age, defined as the time elapsed since its last major repair or replacement. These variables are fundamental for assessing the current structural condition and projecting future maintenance requirements. DS values are typically derived from systematic assessments of individual structural elements based on standardized inspection criteria, such as the five-level DS rating scale, which reflects the severity of material defects and the extent of cross-sectional or surface area loss. Together, these state indicators provide a robust foundation for modelling bridge condition dynamics and informing effective maintenance decision-making.

5.2.2. Maintenance action space

The decision-making process for maintenance interventions was structured through a discrete action space designed to enable systematic and interpretable strategies. For each bridge component, the agent can choose from a standardized set of four distinct maintenance actions. When the agent selects an action, the environment applies deterministic changes to the corresponding component's state:

1. No Action signifies deferring active maintenance, allowing components to continue operating under current conditions subject to ongoing deterioration and the component's state is only affected by the stochastic deterioration model for that time step. No Action signifies deferring active maintenance. This option enables cost-effective strategies by avoiding unnecessary interventions when structural performance remains acceptable.
2. Minor Repair actions address moderate deterioration levels through limited restoration. They reduce the component's DS by one level without significantly altering its age or long-term service life trajectory. These interventions represent routine, low-cost maintenance activities managing localized damage or slowing deterioration progression.
3. Major Repair actions provide substantial restoration capabilities. They improve the component's condition by reducing its DS by two levels while also rejuvenating the component by reducing its effective age by a predefined period which is considered fixed across all components for simplicity. This action extends service life through more comprehensive intervention.
4. Replacement is the most intensive intervention, completely renewing a component by restoring its DS to "*Healthy*" (DS = 0) and resetting its age to zero. Unlike traditional rule-based models that explicitly mask or restrict this action to advanced DS, the proposed framework allows the agent unrestricted access to all actions. Instead, the agent learns to avoid premature replacements through strict economic "overkill" penalties, fostering a robust and naturally learned decision-making process.

5.2.3. Environment transition dynamics

Component deterioration is modelled as a probabilistic process. The probability of a component's DS increasing during a time step is determined by a hazard rate λ calculated as:

$$\lambda = \beta_0 + c_1 \cdot CF + c_2 \cdot \frac{Age}{Age_{max}} \tag{10}$$

where β_0 is the base deterioration rate dependent on the component's MV. c_1 and c_2 are weighting coefficients. In this study, to accurately capture the accelerating degradation of aging infrastructure, the causal factor coefficient c_1, and the age coefficient c_2 are used to ensure that older bridges deteriorate faster, forcing the agent to intervene strategically in later life stages.

The systemic risk of collapse is recalculated after each action. It is derived from the failure probability of individual components, which is a function of their current DS, MV, and CF. These individual probabilities are aggregated using predefined IF that reflect each component's criticality to overall structural stability, providing a direct feedback signal for the risk-minimization objective.

The systemic risk of collapse is recalculated at each time step to provide a direct feedback signal for one of the primary optimization objectives. This is a two-step process that first evaluates the failure probability of individual components and then aggregates them to determine the overall system collapse probability.

First, the failure probability for each individual component i, denoted as $P(F_i)$, is calculated based on its current condition. This model incorporates a base probability derived from its damage state, which is then amplified by multipliers for material vulnerability and causal factors:

$$P(F_i) = \min\left[P_{base}(DS_i) \times (1 + c_{MV}(MV_i - MV_{min})) \times \left(1 + \frac{(CF_i - CF_{min})}{(CF_{max} - CF_{min})}\right), P_{max}\right] \tag{11}$$

where DS_i, MV_i. CF_i are the damage state, material vulnerability index, and causal factor for component i. $P_{base}(DS_i)$ is the base failure probability corresponding to the component's damage state. c_{MV} is the material vulnerability risk coefficient. MV_{min}, CF_{min}, and CF_{max} are normalization constants. P_{max} is a predefined cap on the maximum failure probability for any single component. While Equations (10) and (11) are simplified probabilistic models created to facilitate the extensive simulation required for RL training, their structures are not arbitrary. They incorporate established engineering variables, such as MV and CF, which are widely used in existing infrastructure management literature to model deterioration rates and failure probabilities (Rashidi et al., 2016a, 2016b; Valenzuela et al., 2010).This ensures a logical and engineering-consistent environment for the agent to learn optimal maintenance policies.

Second, these individual probabilities are aggregated to compute the system-level collapse probability, $P(Collapse)$, using a weighted model based on each component's importance:

$$P(Collapse) = 1 - \prod_{i=1}^{n} (1 - w_i\, P(F_i)) \tag{12}$$

where n is the total number of components and w_i is the normalized IF for component i. The final risk component for the reward vector is the negative of $P(Collapse)$, thus incentivizing the agent to select actions that maintain or reduce the probability of systemic failure.

5.2.4. Multi-objective reward structure and utility integration

The agent's learning is guided by a multi-objective reward signal designed to balance the competing goals of bridge maintenance. At each time step t the agent receives an immediate reward vector $r_t = [r_{cost}, r_{risk}, r_{serv}]$, which is calculated after a maintenance action is executed but before the application of stochastic deterioration models. This ensures that the reward directly reflects the consequence of the agent's decision and unfair penalties from random deterioration events is avoided.

- **The Cost Objective** (r_{cost}) represents the direct financial expenditure associated with maintenance interventions. It incorporates the base costs of individual actions along with a fixed mobilization fee. This fee is applied once per time step whenever any maintenance occurs, which inherently incentivizes the agent to consolidate repairs across multiple components rather than performing isolated interventions year after year. Additionally, explicit economic penalties are assigned for disproportionate actions, such as executing major structural repairs on components with only minor damage or intervening on fully healthy elements. These economic constraints ensure that the agent learns an optimal, proportionate maintenance strategy without relying on rigid, pre-programmed rules.
- **The Risk Objective** (r_{risk}) models the expected consequence of failure rather than a raw probability. A tolerable risk threshold τ is established to represent acceptable operational safety margins. If the system-level probability of collapse $P(Collapse)$, exceeded τ, the agent incurs a severe penalty proportional to an expected failure consequence value C_{fail}. The risk reward is formulated as:

$$r_{risk} = -max(0, P(Collapse) - \tau) \times C_{fail} \tag{13}$$

- **The Serviceability Objective** (r_{serv}) is designed to maintain the bridge components within acceptable operational conditions by strictly penalizing severe structural degradation. Rather than rewarding maintenance actions directly, this objective evaluates the ongoing physical state of the infrastructure. Components that remain in healthy, minor, or moderate DS operate without penalty. Conversely, if a component is allowed to deteriorate to a severe or critical DS, the system imposes a substantial financial penalty. This penalty is mathematically scaled by the component's IF, ensuring that the agent learns to prioritize the preservation and timely repair of the bridge's most structurally critical elements.

To resolve the multi-objective signal into a single learning target, a fixed weighted linear sum utility function is employed. Because all objectives are formulated as negative penalties, a linear scalarization provides consistent gradients and effectively captures the trade-offs between cost, risk, and serviceability without the need for complex logarithmic transformations. At any given time-step t, the utility function u applied to the immediate reward vector $r_t = [r_{cost}, r_{risk}, r_{serv}]$ is formulated as:

$$u(r_t) = w_{cost}\left(\frac{r_{cost}}{S_{cost}}\right) + w_{risk}\left(\frac{r_{risk}}{S_{risk}}\right) + w_{serv}\left(\frac{r_{serv}}{S_{serv}}\right) \tag{14}$$

where w_{cost}, w_{risk}, and w_{serv} represent the relative preference weights assigned to the cost, risk, and serviceability objectives, respectively. To prevent any single dimension with naturally larger numerical values from dominating the learning process, the raw penalties are divided by their corresponding normalization scaling factors (S_{cost}, S_{risk}, and S_{serv}).

In the context of sequential decision-making, the RL agent does not merely optimize for the immediate step. Instead, it seeks to maximize the expected cumulative utility over the entire planning horizon. This is defined by the expected scalarized return G_t, which incorporates a discount factor $\gamma \in [0, 1]$ to balance the importance of immediate versus future consequences:

$$G_t = \sum_{k=0}^{\infty} \gamma^k u(r_{t+k}) \tag{15}$$

This linear and discounted formulation ensures a stable optimization landscape, guiding the agent to learn long-term, proactive maintenance strategies while allowing decision-makers to easily encode strategic priorities by adjusting the preference weights.

5.3. Multi-Objective Deep Centralized Multi-Agent Actor-Critic algorithm

The MODRL framework in this study aims to address the complex bridge maintenance optimization challenges, where multiple competing objectives must be balanced simultaneously. The framework adapts the

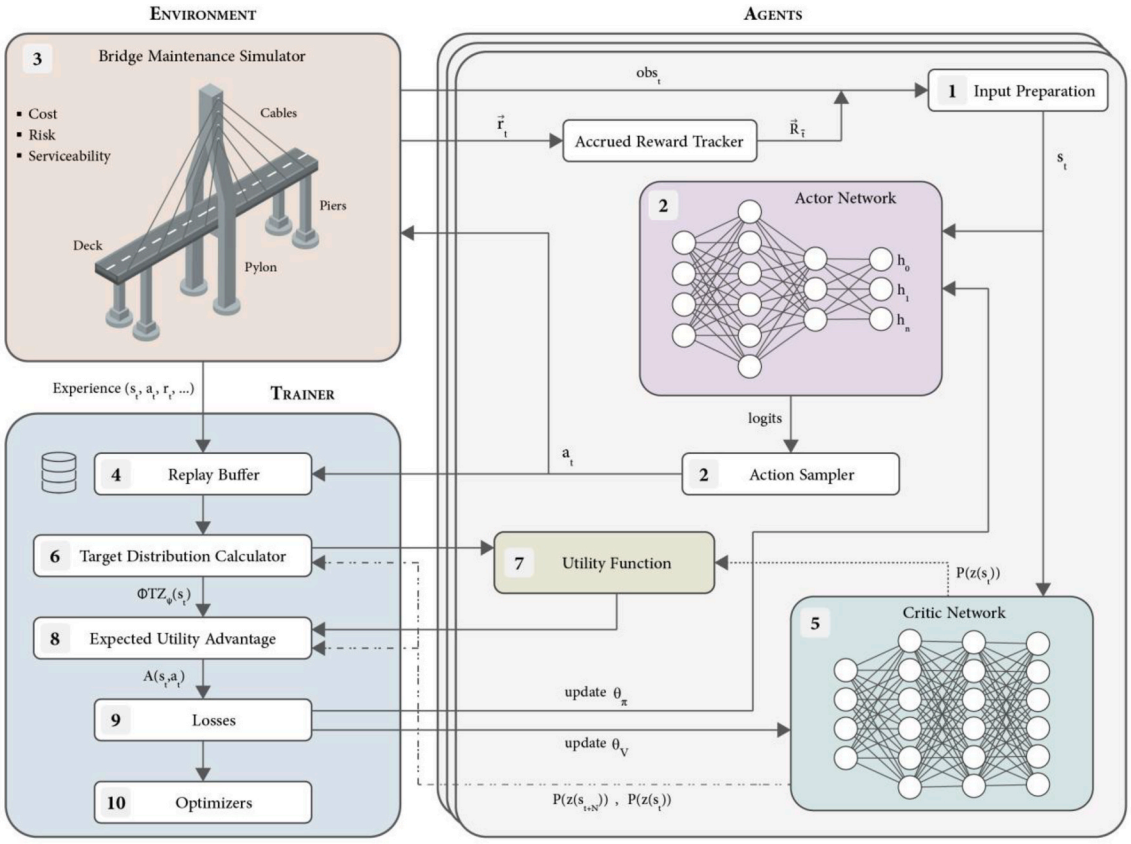


Fig. 2. The architecture of MO-DCMAC algorithm.

MO-DCMAC algorithm, originally proposed by van Remmerden et al. (2025). This algorithm serves as the core decision-making engine of the RL-DSS, extending traditional actor-critic architectures to represent vector-valued reward signals rather than scalar rewards, which enables learning coherent policies that optimally balance competing objectives through utility functions. Despite the implementation by van Remmerden et al. (2025) which is limited to two objectives, our proposed method expands the capability of MO-DCMAC to simultaneously optimize three distinct objectives, thereby enhancing its applicability to more complex, real-world maintenance scenarios. However, incorporating an additional objective increases the dimensionality of the learned return distribution, intensifies computational complexity, and complicates policy optimization under utility-based aggregation. It also heightens sensitivity to reward scaling and weighting, necessitating careful normalization to prevent dominance among objectives. Accordingly, architectural adjustments and training refinements were implemented in the proposed framework to ensure stable convergence and meaningful policy learning within the expanded multi-objective setting.

Specifically, extending the framework from two to three objectives introduces a significant computational challenge related to the curse of dimensionality. Because the distributional critic learns a joint probability mass over the objectives, the output dimension scales exponentially as $\mathcal{N}^d$, where $\mathcal{N}$ is the number of atoms (51 in this study) and d is the number of objectives. Moving from two to three objectives expands the joint atom grid from 2601 to 132,651 parameters per network output.

To ensure computational efficiency and optimization stability at this high-dimensional scale, targeted implementation modifications were required. First, a weighted linear utility function is utilized instead of a non-linear formulation. Because all objectives are formulated as negative penalties, a linear sum provides more consistent gradients during training. Under this fixed linear utility function, the SER and ESR formulations are mathematically equivalent. While this linearity avoids the invalidation of the Bellman equation associated with non-linear utilities (Hayes et al., 2022b), it renders the optimization objective identical to standard scalarized expected-return RL. Consequently, the vector-valued distributional critic is retained for its practical and diagnostic advantages rather than as a fundamentally distinct mathematical objective, as it ensures that cost, risk, and serviceability profiles remain independently observable throughout training and evaluation.

While extending the linear scalarization to three objectives is mathematically straightforward, the core contribution of this work lies in the computational implementation that makes this setting feasible. The distributional Bellman projection, which maps target distributions onto the fixed grid, was entirely vectorized. The original base framework relied on nested iterative loops over the multi-dimensional grid, which becomes computationally inefficient for three objectives. To resolve this, the multi-dimensional target projection was flattened, and probability mass was distributed to the bounding corners simultaneously across the batch using optimized tensor memory scatter operations. This implementation-level transformation shifts the computational complexity onto parallelized C++ and CUDA backend operations. As a result, the framework efficiently handles the increased computational burden of the action-value distribution, reducing training time while maintaining scalability and preserving multi-objective fidelity.

The framework architecture comprises two primary components: autonomous agents that embody the actor-critic structure and interact with the environment, and a centralized trainer that manages the learning process using experiences gathered by the distributed agents. All agents share a common policy and value network backbone, with separate action heads per element. This parameter sharing improves scalability and reduces computational cost while still allowing component-specific decisions. The complete framework architecture and information flow between these components is illustrated in Fig. 2.

The process initiates with input preparation in step (1). For a given time-step, the raw observation from the environment obs_t is combined with the episode's accrued reward vector $\vec{R}_{\tilde{t}}$ to form a comprehensive state representation s_t. Before concatenation, the accrued reward vector $\vec{R}_{\tilde{t}}$ is element-wise normalized by the utility scales to keep magnitudes comparable and prevent any objective from dominating the input. This state vector to the agent's neural networks. The s_t serves as the primary input and is fed into the actor network in step (2), which is responsible for policymaking. This network features initial shared layers for global feature extraction followed by distinct output heads h_n for each bridge component, enabling decentralized action selection within a centralized learning framework. The network processes the state and generates logits, which represent the un-normalized probabilities for each possible maintenance action. An action sampler then uses these logits to select a discrete maintenance action a_t, for each bridge component.

The selected action a_t is executed within the bridge maintenance simulator in step (3), which represents the environment. The simulator evaluates the impact of the action and transitions to a subsequent state. In response, the environment provides a multi-objective reward vector $\vec{r}_t$, capturing the immediate outcomes in terms of cost, risk, and serviceability, along with the next raw observation. The reward vector $\vec{r}_t$ is utilized by the accrued reward tracker to update the cumulative reward total for the episode. The complete transition, denoted as experience, is systematically stored in a replay buffer in step (4) for subsequent training.

The training phase follows an N-step update mechanism, where sequences of transitions are sampled from the replay buffer. The complete operational sequence, detailing both the environment interactions (Fig. 2) and this mathematical training phase, is systematically summarized in Algorithm 1. A key implementation feature in the MO-DCMAC framework is the use of a distributional critic. The Critic Network in step (5) receives the state s_t from the sampled batch and, instead of predicting a single expected value, it learns a discretised probability distribution over the possible future multi-objective returns. It outputs $Z_\psi\left(\vec{z}_t \middle| s_t\right)$, the probability of each possible vectorial return atom $\vec{z}_t$ occurring from s_t. This can be formally expressed as:

$$V(s_t) = \sum_{i \in [\{0,\ldots,\mathscr{N}-1\}]^d} \vec{z}_i Z_\psi\left(\vec{z}_t \middle| s_t\right) \tag{16}$$

where $\mathscr{N}$ is the number of atoms per objective and d is the number of objectives. Instead of predicting a single expected value, the critic represents the full distribution of possible future rewards (the return). This distribution is approximated using $\mathscr{N}$ evenly spaced atoms, where each atom serves as a marker on a grid of possible outcomes. Together, these atoms form a discrete approximation of the return distribution. In parallel, the target distribution calculator in step (6) computes a target distribution by performing a distributional Bellman update on the atoms and projecting the result back onto the critic's support. This creates a stable learning target $\Phi\widehat{T}Z_\psi(s_t)$ for the critic. To guide the policy update, an advantage signal is computed. The expected utility advantage in step (8) module calculates the advantage signal by applying a utility function

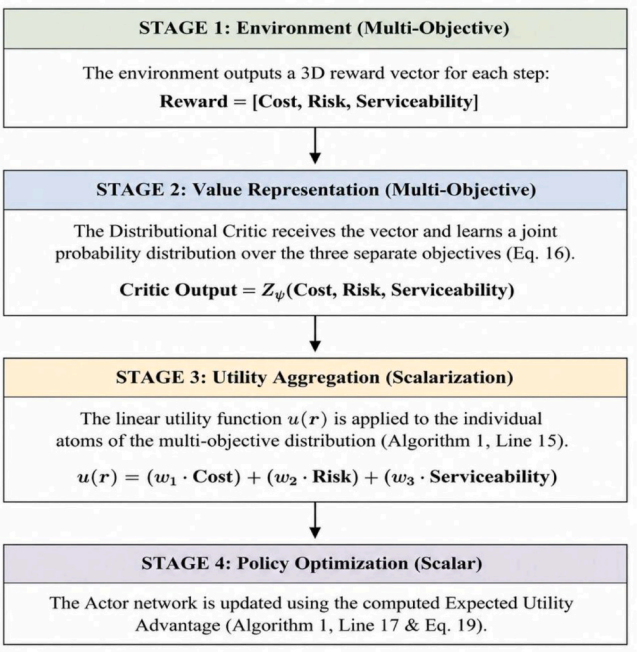


Fig. 3. Four-stage processing pipeline of the MO-DCMAC framework.

in step (7) to the rewards represented in the distributions' supports. The advantage $A(s_t, a_t)$ is the difference between the expected utility of the n-step return and the expected utility of the current state's value distribution:

$$A(s_t,a_t) = \left(\sum_i u\left(\vec{R}_{\tilde{t}} + \gamma^n\left(\vec{r}_t + \gamma\vec{z}_t\right)\right) Z_\psi\left(\vec{z}_i \middle| s_{t+1}\right)\right) - \left(\sum_i u\left(\vec{R}_{\tilde{t}} + \vec{z}_i\right) Z_\psi\left(\vec{z}_i \middle| s_t\right)\right) \tag{17}$$

This utility-based advantage provides a gradient signal encouraging action that leads to higher expected utility outcomes. The computed advantage and distributions are then passed to the losses module in step (9). The critic's value loss $\mathscr{L}^V$ is the Kullback-Leibler (KL) divergence between the target distribution and the critic's predicted distribution, optimized via its cross-entropy component:

$$\mathscr{L}_V = D_{KL}(\Phi\widehat{T}Z_\psi(s_t) \| Z_\psi(s_t)) \tag{18}$$

The actor's policy loss $\mathscr{L}^\pi$ uses the calculated advantage to update the policy parameters. Given the MA structure with a factorized action space, the loss is summed over all components and the global action. The policy is conditioned on the comprehensive state representation s_t, which concatenates the raw environmental observations with the normalized accrued reward:

$$\mathscr{L}_\pi = -\left(\sum_{j=0}^{N-1} \log\left(\pi_\theta^{c_j}\left(a_t^{c_j} | s_t\right)\right)\right)\widehat{A}(s_t,a_t) \tag{19}$$

where $\widehat{A}(s_t, a_t)$ is the normalized advantage calculated from s_t sampled from the replay buffer. c_j is the component. Finally, these losses are combined with an entropy bonus $H(\pi)$ to encourage exploration and passed to the optimizers in step (10), which perform gradient descent to update θ_V and θ_π, the parameters of the critic and actor networks, respectively. This iterative process refines the agent's policy, progressively enhancing its ability to make optimal maintenance decisions that balance the conflicting objectives. Algorithm 1 outlines the complete

MO-DCMAC training loop, clarifying the integration of the N-step target projection, advantage computation, and multi-objective loss aggregation.

Algorithm 1. MO-DCMAC Training Loop

Input: Bridge environment; Actor π_{θ}; Distributional Critic Z_{ψ}; Utility function $u(\cdot)$
Output: Optimized network parameters $\boldsymbol{\theta}, \boldsymbol{\psi}$

Initialize π_{θ}, Z_{ψ}, replay buffer $\mathscr{B}$, and joint atom grid $\boldsymbol{z}$
for each environment step $t = 1, \ldots, T_{max}$ **do**
  //1. Environment Interaction
  Formulate state $\boldsymbol{s}_t \leftarrow [\boldsymbol{obs}_t \| \overline{\boldsymbol{R}}_t \oslash \boldsymbol{S}]$ ▷ Include normalized accrued reward
  Sample joint action $\boldsymbol{a}_t \sim \pi_{\theta}(\cdot|\boldsymbol{s}_t)$
  Execute $\boldsymbol{a}_t$, observe next observation $\boldsymbol{obs}_{t+1}$, reward $\boldsymbol{r}_t$, and done flag d_t
  Update accrued reward: $\overline{\boldsymbol{R}}_{t+1} \leftarrow \overline{\boldsymbol{R}}_t + \boldsymbol{r}_t$
  Store transition $(\boldsymbol{s}_t, \boldsymbol{a}_t, \boldsymbol{r}_t, \overline{\boldsymbol{R}}_t, \boldsymbol{s}_{t+1}, d_t)$ in $\mathscr{B}$
  if time to update **and** $|\mathscr{B}| \geq N$ **then**
    Sample the recent sequence of N transitions from $\mathscr{B}$
    //2. Target Projection
    Compute N-step return distributions via Bellman recursion
    Project returns onto fixed grid $\boldsymbol{z}$ to obtain target distributions $\Phi\widehat{T}Z_{\psi}(\boldsymbol{s}_t)$
    //3. Advantage Computation (Eq. (17))
    Compute expected utility of current state: $V_u(\boldsymbol{s}) \leftarrow \sum_i u(\overline{\boldsymbol{R}} + \boldsymbol{z}_i) \cdot Z_{\psi}(\boldsymbol{z}_i|\boldsymbol{s})$
    Compute expected utility of target: $Q_u(\boldsymbol{s}, \boldsymbol{a}) \leftarrow \sum_i u(\overline{\boldsymbol{R}} + \boldsymbol{z}_i) \cdot \Phi\widehat{T}Z_{\psi}(\boldsymbol{z}_i|\boldsymbol{s})$
    Calculate and normalize advantages: $\widehat{A} \leftarrow Normalize(Q_u - V_u)$
    //4. Loss Aggregation & Optimization
    Critic Loss: $\mathscr{L}_V \leftarrow D_{KL}(\Phi\widehat{T}Z_{\psi}(\boldsymbol{s}) \| Z_{\psi}(\boldsymbol{s}))$ (Eq. (18))
    Actor Loss: $\mathscr{L}_{\pi} \leftarrow \sum_j \log \pi_{\theta^{(j)}}(a^{(j)}|\boldsymbol{s}) \cdot \widehat{A}$ (Eq. (19))
    Total Loss: $\mathscr{L} \leftarrow \mathscr{L}_{\pi} + c_V \mathscr{L}_V - c_H(t)\mathscr{H}(\pi_{\theta})$
    Perform gradient descent step on $\mathscr{L}$ to update $\boldsymbol{\theta}, \boldsymbol{\psi}$
  end if
  if dt is True **then**
    Reset environment and accrued reward ($\overline{\boldsymbol{R}} \leftarrow 0$)
  end if
end for

Notation. $\overline{\boldsymbol{R}}_t$: cumulative per-objective reward vector; $\boldsymbol{S} = [S_{cost}, S_{risk}, S_{serv}]$: normalization scales; $\boldsymbol{z}_i$: joint atom vector; $\Phi\widehat{T}Z_{\psi}$: projected target distribution; c_V, $c_H(t)$: critic and (decaying) entropy coefficients; d_t: Boolean episode termination (done) flag; $\oslash$: element-wise division.

The overall processing pipeline of the proposed MO-DCMAC framework is illustrated in Fig. 3. In Stage 1, the environment generates a vector-valued reward signal composed of maintenance cost, structural risk, and serviceability, as explained in Section 5.2.4. In Stage 2, the distributional critic learns a joint probability distribution over these objectives through the multi-objective Bellman update. In Stage 3, the weighted linear utility function defined in Eq. (14) is applied to the learned return distribution to obtain a scalarized utility representation, while the corresponding expected scalarized return is computed according to Eq. (15). Finally, in Stage 4, the actor network is updated using the expected utility advantage and the policy optimization procedure. The figure summarizes how the proposed framework integrates multi-objective return modelling, utility-based scalarization, and policy optimization within a unified learning architecture.

While Fig. 3 depicts a late-scalarization pipeline, as established earlier in this section, this architecture does not alter the fundamental optimization objective under fixed linear utility function. The vector-valued distributional critic is retained not out of theoretical necessity, but for its practical value in preserving per-objective transparency, enabling independent monitoring of cost, risk, and serviceability throughout training and within the DT-based management framework.

5.4. Training and implementation details

The MO-DCMAC model was implemented in Python v3.8 using the PyTorch v2.0, with training and evaluation on a PC equipped with an NVIDIA RTX 2080 Ti GPU and 128 GB of RAM.

Table 5
Hyper-parameters used for the MO-DCMAC.

Hyper-parameters	Value
Discount Factor (γ)	0.995
Training Steps	3×10^7
N-step Update	256
Replay Buffer Size	4096
Episode Length	100
Actor Learning Rate	2e-4 → 2e-5
Critic Learning Rate	2e-3 → 2e-4
Gradient Clipping Norm	2.0
Value Function Coefficient	0.5
Entropy Coefficient	0.1
Critic Atoms per Objective	51
Critic Value Range (Cost)	[-50000, 0]
Critic Value Range (Risk)	[-80000, 0]
Critic Value Range (Serviceability)	[-50000, 0]
Utility Weight (Cost)	1.10
Utility Weight (Risk)	1.00
Utility Weight (Serviceability)	0.95

The training process was executed within a structured framework managing hyper-parameter configuration, environment setup, agent initialization, and operational loops for training and evaluation. Hyper-parameters were iteratively refined through systematic exploration to balance learning stability with bridge maintenance task performance. The key hyper-parameters employed in this study are detailed in Table 5, with value ranges chosen through a combination of empirical tuning during preliminary experiments and reference to values commonly used in related studies.

In addition to the algorithmic hyper-parameters, the simulation environment was calibrated to reflect realistic economic constraints and physical deterioration dynamics. Table 6 summarizes these core environmental parameters, establishing the penalty-based framework that guides the agent's learning process. Notably, the environment incorporates a fixed mobilization cost to encourage clustered maintenance, alongside specific financial penalties for wasteful or overkill interventions. This configuration ensures the agent learns a proportionate, economically viable maintenance strategy. Furthermore, while these parameters were selected by the authors for the purpose of methodological demonstration, practical implementation would require systematic stakeholder engagement. In real-world applications,

Table 6
Economic, risk, and deterioration parameters for the simulation environment.

Parameters	Value	Description
Action costs	0, 150, 500, 2000	Base financial cost for [No Action, Minor Repair, Major Repair, Replace].
Mobilization cost	600	Fixed setup fee applied per time step if any maintenance occurs.
Major repair age reduction	20 years	Structural age reduction following a major repair.
Wasteful and overkill penalties	200 to 300	Fines for over-maintaining healthy or lightly damaged elements.
Serviceability penalties	400, 2000	Fines for components reaching severe (DS = 3) or critical (DS = 4) states.
Tolerable risk threshold (τ)	0.02	Maximum acceptable system-level probability of collapse.
Consequence of failure (C_{fail})	80000	Penalty applied if the tolerable risk threshold is exceeded.
Base failure probability (P_{base})	5×10-4, 2×10-3, 0.01, 0.05, 0.20	Component failure probabilities corresponding to DS levels 0 through 4.
Maximum failure probability (P_{max})	0.95	Absolute cap on the calculated failure probability for any single component.
Hazard coefficient (c_1)	0.16	Weighting of CFs on the deterioration rate.
Hazard coefficient (c_2)	0.10	Weighting of structural age on the deterioration rate.

preferences would be elicited through structured surveys or semi-structured interviews with asset managers and infrastructure owners to capture their priorities.

The MO-DCMAC framework supports domain randomization to train across diverse bridge scenarios by varying parameters like component age, DS, MV, R, and EA, allowing the agent to develop generalized maintenance policies adaptable to multiple bridge types and conditions. However, for this study, domain randomization was not employed. Instead, dedicated agents were trained specifically for each case study, with simulation environments configured to match the known attributes of each bridge type. This approach develops tailored optimized maintenance policies for the specific target bridge, avoids generalization-induced performance dilution, and reduces computational costs compared to learning across diverse scenarios.

The agent was trained over a set of environment steps, with each training episode simulating a 100-year bridge lifecycle where time-steps represent annual periods. Future rewards were discounted using a gamma factor to modulate long-term planning horizons. Learning updates occurred after collecting fixed numbers of environment interactions, with transitions stored in replay buffers containing processed observations, selected actions, multi-objective reward vectors, subsequent observations, terminal flags, and discount factors.

Neural network optimization employed Adam optimizers with learning rates linearly decayed from initial to final values throughout training. Gradient clipping was applied to prevent excessive parameter updates that could destabilize learning processes. The composite loss function incorporated weighted critic losses with entropy bonuses added to actor losses for exploration encouragement, subject to coefficient decay during training progression.

The training loop involved repeated environment resets allowing agent interactions through policy-based action selection and environment response observation through state transitions and reward vectors. Interactions were systematically logged with learning updates performed after designated step intervals. Training progress including actor-critic losses, accumulated penalties, and overall linear utility values was continuously monitored through logging systems.

6. Case studies

The proposed framework was evaluated through three distinct bridge maintenance scenarios, each representing different structural characteristics, environmental conditions, and maintenance challenges. These case studies were designed to demonstrate the model's adaptability and effectiveness across diverse bridge configurations while maintaining a consistent evaluation framework. The evaluation focused on the model's ability to generate maintenance policies that balance the multi-objective trade-offs of minimizing costs, reducing collapse risks, and maximizing performance over a 100-year planning horizon, in line with standard service life in AS 5100:2017 (Australia) (Australian Standard, 2004) and AASHTO's guidelines (Hopper et al., 2022).

Three bridge structures were selected for evaluation, including Werrington bridge, Mt Emu Creek bridge, and McKanes Falls bridge with different types of materials, age and environmental conditions. Table 7 presents the specific characteristics of each bridge.

As described in Section 5.4, while the MO-DCMAC framework incorporates capabilities for generalized training via domain randomization, a specific training approach was deliberately employed for this study. Dedicated agents were trained for each of the three case studies, with the simulation environment configured to precisely match the known attributes of their corresponding bridge. This approach allows the framework to learn a highly optimized policy for each well-defined scenario. Training these dedicated agents was computationally intensive, requiring an average of 33 h per case study on the hardware specified in Section 5.4. Furthermore, to ensure a consistent basis for multi-objective evaluation across all scenarios, the normalization scaling factors for the linear utility function were uniformly parameterized for all three agents. The cost and serviceability scaling factors (S_{cost} and S_{serv}) were set to 15,000, while the risk scaling factor (S_{risk}) was set to 8000. These values were selected to normalize the naturally larger numerical penalty values into a comparable range, preventing any single objective from dominating the learning process and fostering policies that strategically balance risk mitigation with cost-efficiency.

To maintain a standardized evaluation framework across all scenarios, the simulation environment was governed by the uniform economic, risk, and deterioration parameters previously detailed in Table 6. The underlying deterioration dynamics were driven by material-specific base deterioration rates (β_0), ranging from 0.015 for precast concrete to 0.050 for timber components, which were further amplified by the causal factor and structural age coefficients (c_1 and c_2). Similarly, the risk model utilized the standardized base failure probabilities, while intervention expenditures were dictated by the fixed action and mobilization costs outlined in the simulation configuration. These values were established as baseline input parameters to generate consistent, interpretable deterioration patterns and realistic economic constraints across all bridge types, rather than being calibrated from specific empirical data.

This experimental design enables comprehensive evaluation of the MO-DCMAC framework's ability to learn effective maintenance policies across diverse bridge types while maintaining consistent optimization objectives and evaluation criteria. The following sections present detailed results and analysis of the learned policies for each scenario.

6.1. Case study I: werrington bridge, NSW Australia

For the first case study, the Werrington bridge, as shown in Fig. 4, is considered to illustrate the RL-DSS framework for maintenance policy decisions. This bridge was built in 1992 over the Great Western Highway and connects the north and south Penrith campuses of Western Sydney University. This 46.2 m cable-stayed bridge has a composite steel-concrete deck supported by an A-shaped steel pylon, three lines of four bearings, and two sets of eight stay cables anchored to the pylon and a southern concrete block.

The training performance of the MO-DCMAC framework for the Werrington bridge scenario is illustrated in Fig. 5(I), tracking the evolution of cumulative cost, risk, and serviceability penalties over 30 million simulation steps. It should be noted that the monetary values and

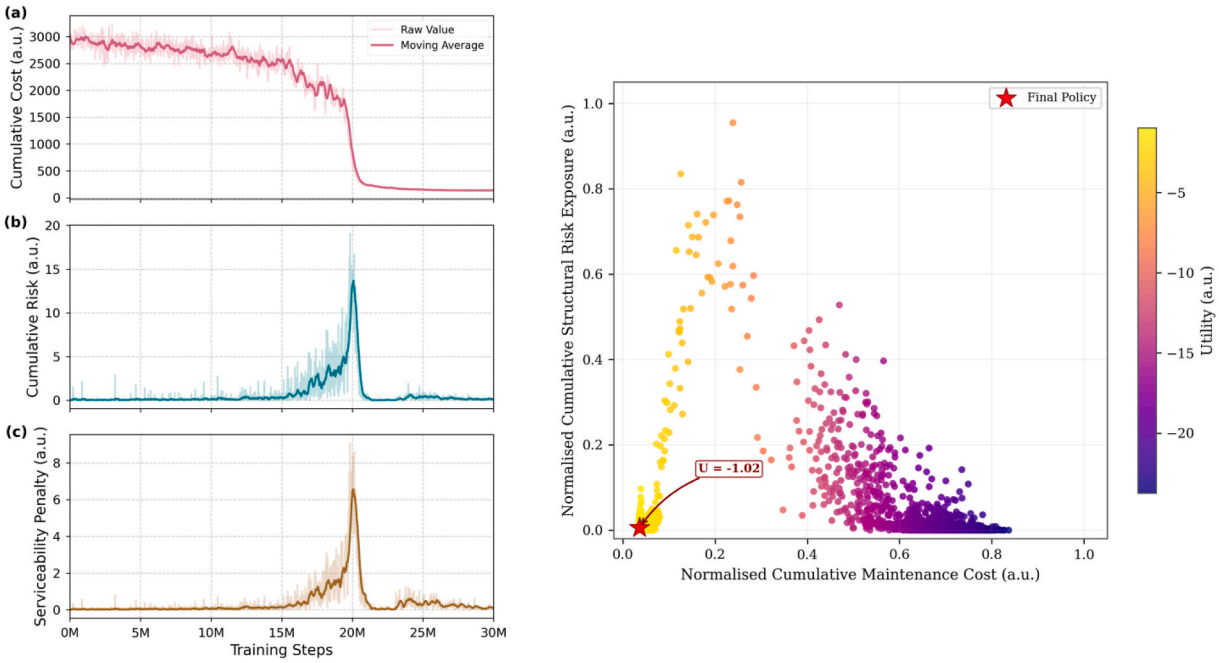
Table 7
Characteristics of each bridge case study and allocated data for bridge element types.

Bridge	Component	Material	IF	DS	Age	Environment	Traffic
Werrington	Deck	Reinforced Concrete	0.6	0	33	Moderate	Local access
	Cable	Steel	0.8	0	33		
	Pylon	Steel	1.0	0	33		
Mt Emu Creek	Deck	Precast Concrete	0.9	2	45	Harsh	Collectors
	Railway	Steel	0.7	0	45		
	Piers	Brick Masonry	1.0	1	45		
McKanes Falls	Deck	Timber	0.7	4	132	Moderate	Local access
	Truss	Timber	0.5	3	132		
	Piers	Reinforced Concrete	1.0	0	39		



(a) (b)

Fig. 4. (a): Werrington bridge, (b) 3D model.



(I) (II)

Fig. 5. Training performance of the MO-DCMAC for the Werrington bridge case study. (I): Evolution of cumulative cost (a), risk (b), and serviceability (c) over 30 million training steps; (II): The policy training trajectory showcasing the training evolution of normalized cost and risk.

penalties reported herein are expressed in arbitrary units (a.u.), corresponding to the synthetic economic parameters established in Section 5 to demonstrate the framework's behavioural logic rather than exact financial forecasting.

As shown in Fig. 5(I)–(a), the cost trajectory demonstrates a clear reduction and successful learning curve. Initially, the agent starts with high cumulative costs due to a randomly initialized policy. As training progresses, the agent gradually learns to minimize unnecessary interventions. However, the learning dynamics for structural risk (Fig. 5 (I)–(b)) and serviceability (Fig. 5(I)–(c)) reveal the complexity of the agent's exploration process. In the early stages, while attempting to aggressively reduce costs by deferring maintenance, the agent allows the bridge components to deteriorate significantly. This triggers the severe cliff-edge penalties embedded in the environment, specifically, the high expected failure cost when the 2.0% system risk threshold is breached, and the strict serviceability fines for components reaching critical damage states (DS 3 and DS 4). These boundary violations manifest as massive spikes in both risk and serviceability penalties between 15 and 20 million training steps. Upon experiencing these catastrophic simulated outcomes, the agent rapidly corrects its behaviour. It learns that while reducing raw maintenance cost is beneficial, failing to prevent severe damage incurs an overwhelming mathematical penalty. Consequently, the policy sharply converges, bringing cost, risk, and serviceability penalties down to near-zero levels. This indicates that the agent has successfully learned to keep the bridge within the strict zero-penalty safety zone through timely, batched, and cost-efficient interventions.

Fig. 5(II) illustrates the policy training trajectory of the MO-DCMAC optimization. Rather than plotting all 30 million individual steps, each point on this scatter plot represents a distinct policy snapshot evaluated periodically during the training process. While the algorithm inherently optimizes three distinct objectives, the plot visualizes the primary engineering trade-off for normalized cumulative maintenance cost versus normalized cumulative structural risk. The distribution clearly highlights the conflicting nature of these objectives. Policies clustered in the bottom-right exhibit high costs but maintain near-zero risk (over-maintenance), whereas points extending toward the top-left represent highly conservative spending strategies that expose the structure to dangerous risk levels (under-maintenance). The final selected policy, marked with a red star ($U = -1.02$), demonstrates a converged, balanced policy state.

The results of 100-year maintenance planning for Werrington bridge are illustrated in Fig. 6. The temporal evolution of DS for critical bridge components, including deck, cables, and pylons is illustrated in Fig. 6(a).

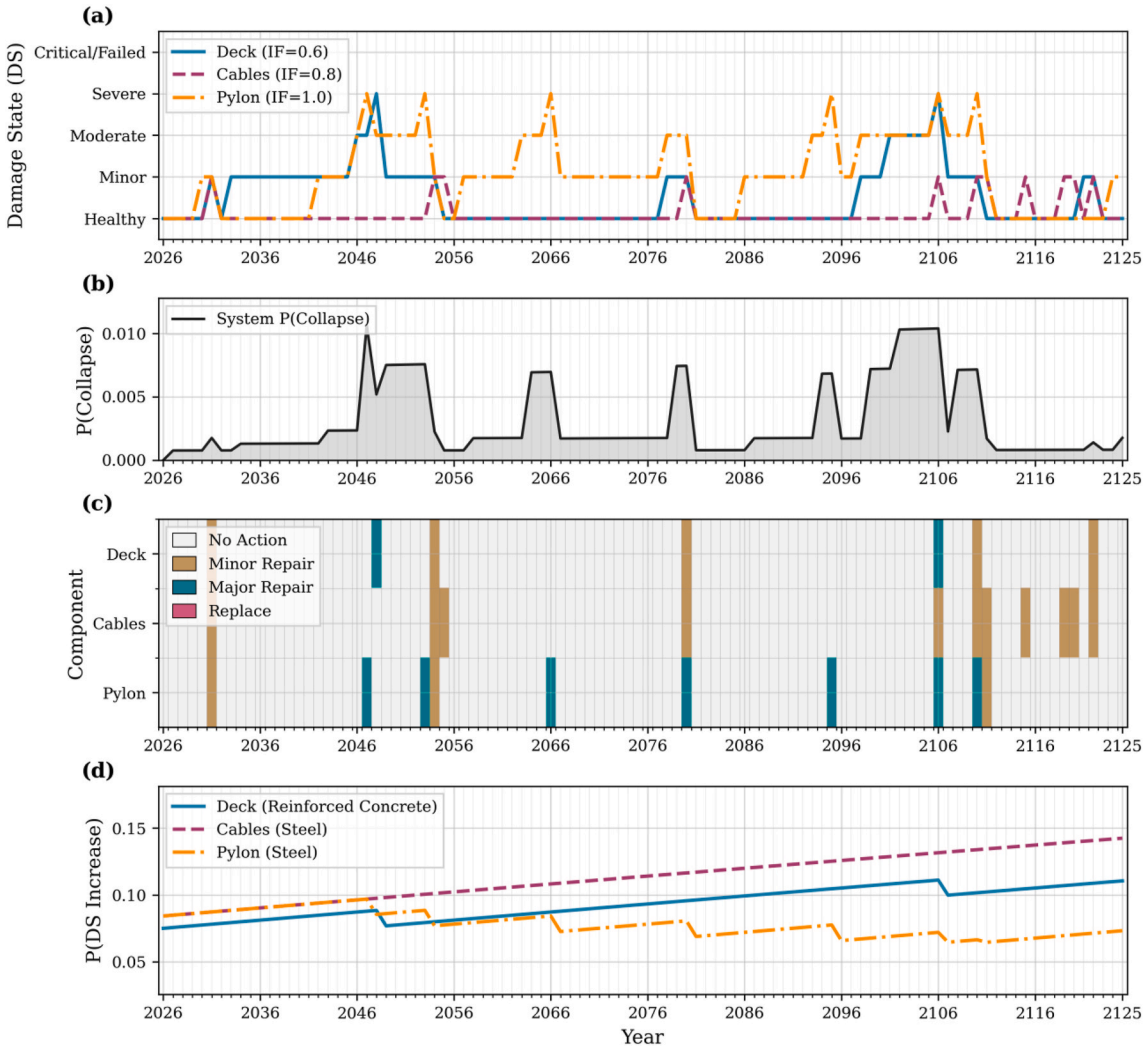


Fig. 6. The results of 100-year maintenance planning for Werrington bridge; (a): Distribution of component DS; (b): System-level probability of collapse; (c): Distribution of maintenance actions by components; (d): The probability of future deterioration.

The degradation patterns reveal that the learned policy does not attempt to maintain all components in a permanently pristine state (DS = 0). Instead, the agent exhibits a patient, controlled-deterioration strategy, allowing components to naturally degrade to minor (DS = 1) or moderate (DS = 2) damage levels. This behaviour perfectly aligns with the environment's serviceability logic, which imposes zero financial penalties for these lower damage states. However, the agent swiftly intervenes before or immediately after a component reaches the severe state (DS = 3) to prevent the heavy fines associated with critical deterioration.

Fig. 6(b) illustrates the time series of the system-level probability of. The overall system risk is maintained at a very low and safe level, generally fluctuating between 0.001 and 0.010. The distinct spikes in risk (e.g., around the years 2047, 2065, and 2106) correspond to moments when multiple components simultaneously reach advanced states of deterioration. Crucially, the maximum risk never exceeds the predefined 0.02 tolerable risk threshold. This demonstrates that the agent has successfully mapped the system's safety boundaries, intervening just in time to avoid the severe expected-failure-cost penalties while avoiding overspending on unnecessary safety margins.

The agent's specific intervention strategy is revealed in Fig. 6(c). The most prominent feature of this policy is action batching. Interventions are heavily clustered in specific years (e.g., 2050, 2080, 2106), where the agent repairs two or more components simultaneously. This synchronized scheduling is a direct, learned response to the fixed mobilization cost. Rather than dispatching crews frequently for isolated repairs, the agent waits until multiple components require attention, thereby minimizing setup overheads. Furthermore, the choice of actions in Fig. 6(c) reflects the influence of component IF and overkill penalties. The pylon, which has the highest structural importance (IF = 1.0), is treated almost exclusively with major repairs to ensure its rapid restoration and age reduction. Conversely, the deck and cables frequently receive cheaper Minor Repairs to patch moderate damage.

Finally, Fig. 6(d) illustrates the underlying annual probability of deterioration or a damage state increase for each component. This metric represents the inherent deterioration risk driven by material properties, age, and environmental conditions. It confirms that the steel components (cables and pylon) have a higher base deterioration rate than the reinforced concrete deck. The distinct drops in these probability curves are a direct consequence of the agent's actions. Interventions such as major repair and replace reduce the effective age of a component, which in turn lowers its probability of future degradation. This provides clear visual evidence that the agent has not only learned to react to the current DS but has also internalized the concept of reducing future risk by resetting a component's effective age. Each major repair pushes the component's deterioration curve down, providing more time until the next intervention and confirming the policy's proactive, long-term nature.



Fig. 7. (a): Mt Emu Creek bridge, (b) 3D model.

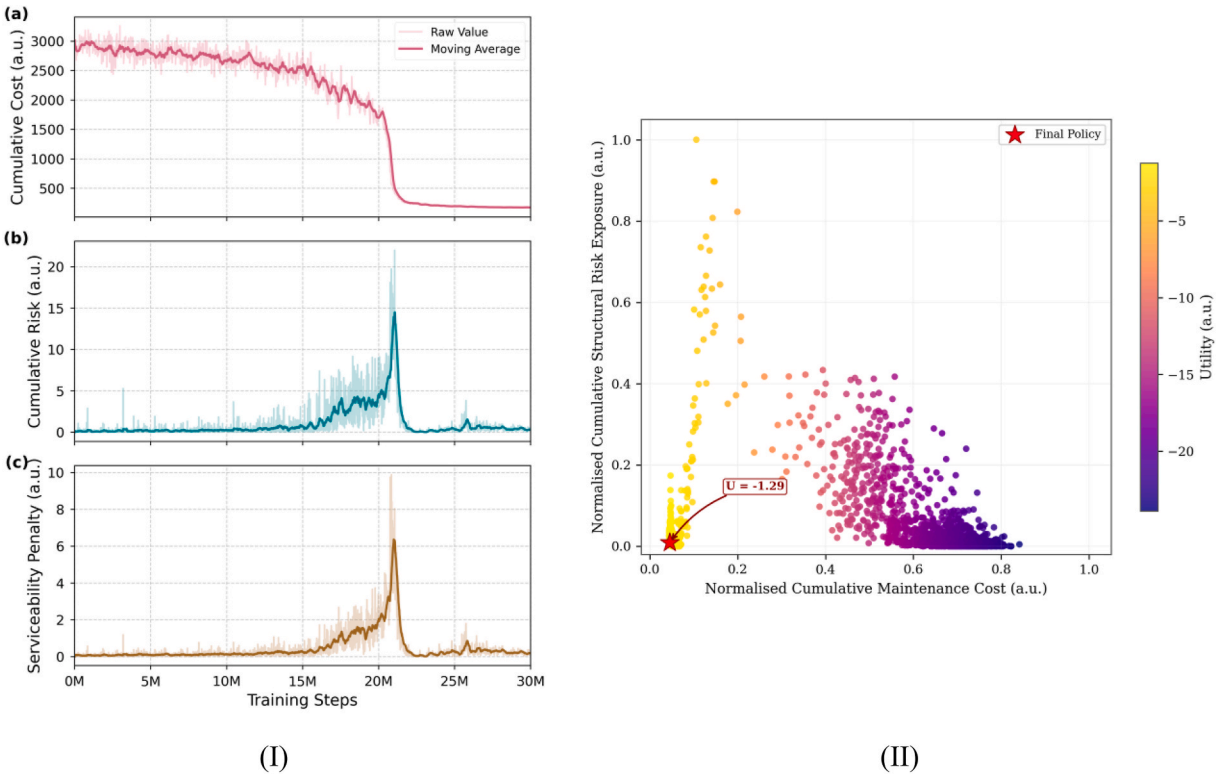


Fig. 8. Training performance of the MO-DCMAC for the Mt Emu Creek bridge case study. (I): Evolution of cumulative cost (a), risk (b), and serviceability (b) over 30 million training steps; (II): The policy training trajectory showcasing the training evolution of normalized cost and risk.

6.2. Case study II: Mt Emu creek bridge, victoria Australia

For the second case study, the Mt Emu Creek bridge, situated near Melbourne, is used to investigate the performance of the proposed approach. This bridge, as shown in Fig. 7, carries a single-track rail line over Mt Emu Creek. Designed around 1980, the bridge comprises five brick piers with concrete headstocks, two concrete abutments protected by boulders, and precast concrete beams.

The training dynamics for the Mt Emu Creek bridge, presented in Fig. 8(I), exhibited a convergence pattern consistent with the Werrington bridge. The agent successfully stabilizes, converging on a cost-efficient policy that drives risk and serviceability penalties back down to near zero. This consistent learning trajectory across different bridge configurations, material properties, and initial conditions underscores the MO-DCMAC robustness in identifying efficient maintenance strategies irrespective of the specific scenario.

Similarly, the training trajectory depicted in Fig. 8(II) confirms the inherent conflict between economic expenditure and structural safety. While the shape of the solution space mirrors the first case study, the final selected policy converges at a slightly lower overall utility ($U = -1.29$) compared to Werrington. This mathematical shift is a direct and logical consequence of the Mt Emu Creek bridge's composition. It features highly vulnerable brick masonry components operating under harsher environmental exposure, as defined in Table 7. Consequently, the agent is forced to absorb slightly higher baseline maintenance costs to keep the structure safe.

Fig. 9 shows the 100-year maintenance plan for Mt Emu Creek bridge. Exposed to harsher environmental conditions and utilizing more vulnerable materials like brick masonry. The resulting policy differs notably from the Werrington case, demonstrating the framework's ability to adapt to distinct asset management challenges.

Fig. 9(a) shows the component DS distribution, and Fig. 9(b) shows the system-level probability of collapse. The agent successfully keeps the system risk low throughout the entire century, never coming close to the tolerable safety threshold. It achieves this strict safety profile by tightly controlling the physical condition of the most critical elements, rarely allowing them to remain in moderate or severe DS for prolonged periods.

The agent's specific intervention strategy is revealed in Fig. 9(c). A striking characteristic of this policy is the distinct treatment of different structural components based on their properties. For the more durable precast concrete deck, the agent applies occasional, batched major repairs. Conversely, for the highly critical and fast-deteriorating brick masonry piers and steel railway, the agent relies almost exclusively on

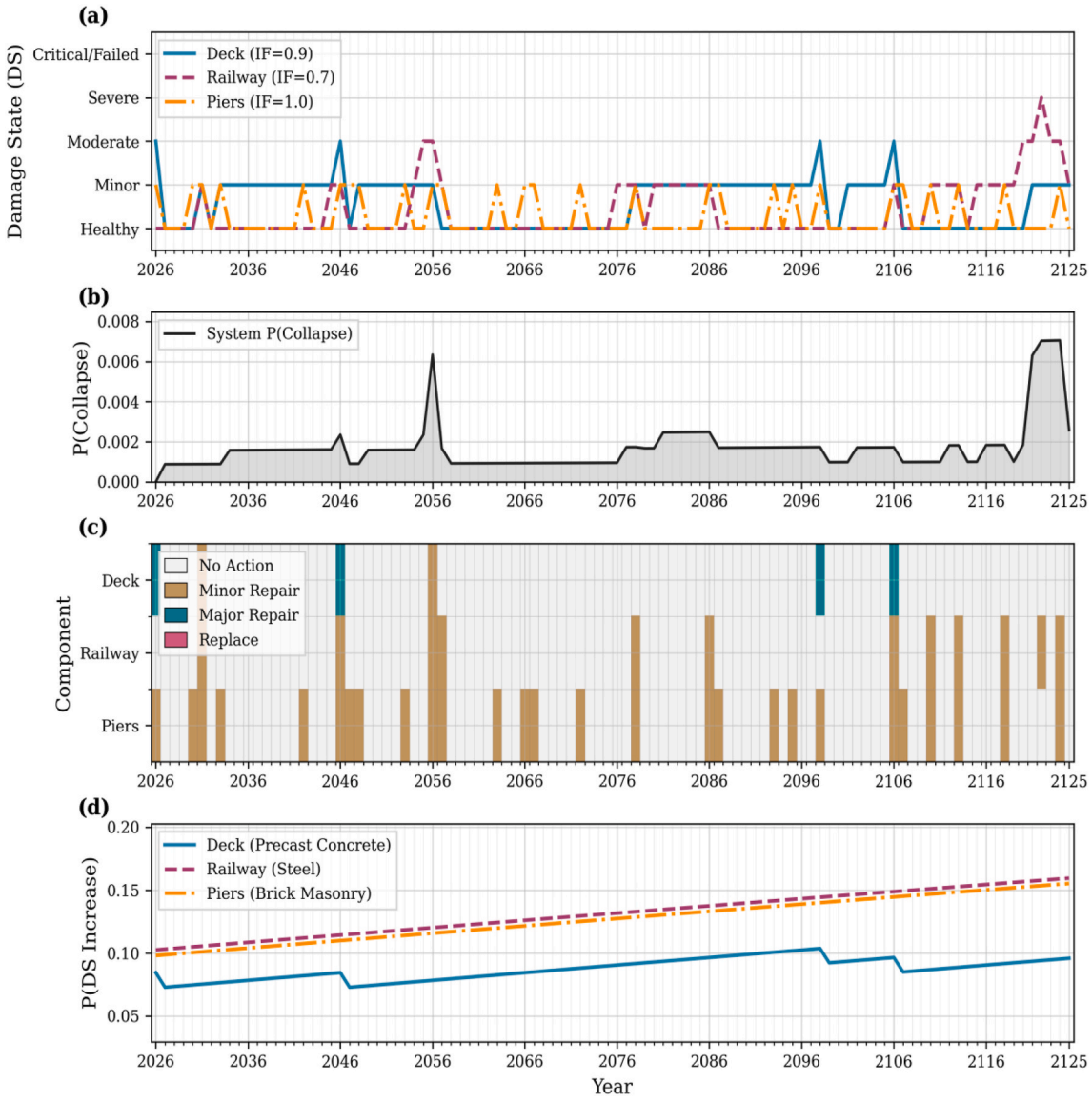


Fig. 9. The results of 100-year maintenance planning for Mt Emu Creek bridge; (a): Distribution of component DS; (b): System-level probability of collapse; (c): Distribution of maintenance actions by components; (d): The probability of future deterioration.

highly frequent minor repairs. This behavior is a mathematically optimized preventative maintenance strategy discovered by the agent. Based on the defined environment rules, the agent calculated that performing frequent, low-cost minor upkeep is significantly cheaper and safer than allowing a component to degrade to a severe state. Waiting for severe damage would force the agent to incur the higher costs of major repairs alongside massive serviceability and structural risk penalties. Furthermore, by applying minor repairs to light damage, the agent avoids the overkill penalties designed to discourage excessive structural interventions. While applying frequent minor repairs represents the optimal mathematical solution for the given constraints, dispatching crews this often may appear logistically inefficient for real-world



(a) (b)

Fig. 10. (a): McKanes Falls Bridge, (b) 3D model.

application. This outcome highlights how sensitive RL agents are to the precise balance of reward weights. In this scenario, the fixed mobilization cost is simply outweighed by the severe financial consequences of allowing structural degradation. In future work, these operational costs and penalty ratios will need further calibration to encourage action batching that more closely aligns with conventional engineering schedules.

Finally, Fig. 9(d) confirms the physical basis for these decisions. It visualizes the annual probability of a DS increase, showing that the brick masonry piers and steel railway have inherently steeper deterioration rates compared to the concrete deck. This justifies the agent's focus and frequent interventions on these specific components, proving the agent has learned to proactively manage the underlying physical decay of the structure.

6.3. Case study III: McKanes Falls Bridge, NSW Australia

A heritage-listed timber truss bridge, the McKanes Falls Bridge (Fig. 10), was selected to investigate the performance of the proposed approach for timber structures. Built in 1893 over the Coxs River, NSW, it has two 27 m timber truss spans on sandstone abutments and a central concrete pier. The 4.5 m-wide deck consists of lateral timber beams on longitudinal girders and has no pedestrian footpaths.

This scenario represents the most demanding test for the agent, starting the 100-year simulation with the bridge already at an advanced age of 132 years and components in critical and severe DS. The training process, shown in Fig. 11(I), reveals both the robustness of the learning algorithm and its ability to navigate complex trade-offs under significant initial stress.

Unlike the previous scenarios, the training dynamics display a massive initial shock at the very beginning of the simulation. Because the bridge starts in a near-failure state, both the risk and serviceability penalties begin at highly elevated levels. The agent is forced into immediate crisis management, rapidly learning to apply stabilizing interventions to bring the system under control. Once the initial crisis is resolved, the training follows the familiar pattern observed in the earlier case studies. The agent undergoes an exploratory phase characterized by a sharp penalty spike before successfully converging to a stable, low-penalty policy. This demonstrates that the framework can successfully rehabilitate and subsequently maintain severely degraded historical assets.

This outcome is further contextualized by the training trajectory shown in Fig. 11(II). The fundamental shape of the trade-off space remains consistent with previous cases. The final converged policy achieves a utility of −1.28, which is highly comparable to the Mt Emu Creek scenario. This reflects the equivalent difficulty of managing a highly vulnerable, aged timber structure. Despite the fragile starting conditions, the agent successfully isolates an optimal policy that balances necessary structural investments with long-term safety.

The 100-year maintenance plan for the McKanes Falls Bridge, presented in Fig. 12, showcases the MO-DCMAC agent's adaptive crisis management capability. Because the bridge begins the simulation in a severely degraded state, the agent is forced to adopt a fundamentally different initial strategy compared to the previous scenarios.

As shown in Fig. 12(a) and (c), the simulation starts with the timber deck in a critical state (DS = 4) and the timber truss in a severe state (DS = 3), both at an advanced age of 132 years. In response, the agent immediately executes a Replace action on both timber components in the very first year. The agent correctly learned that applying a standard major repair to a 132-year-old component would barely reduce its effective age or damage state, leaving the system exposed to ongoing, massive serviceability and risk penalties. Instead, the agent justifies the highest possible upfront financial cost of full replacement to instantly reset the structural age to zero and restore the components to a healthy state.

Following this initial stabilization, the agent transitions into a long-term maintenance regime that resembles the strategies observed in the previous case studies. As seen in the action distribution (Fig. 12(c)), the

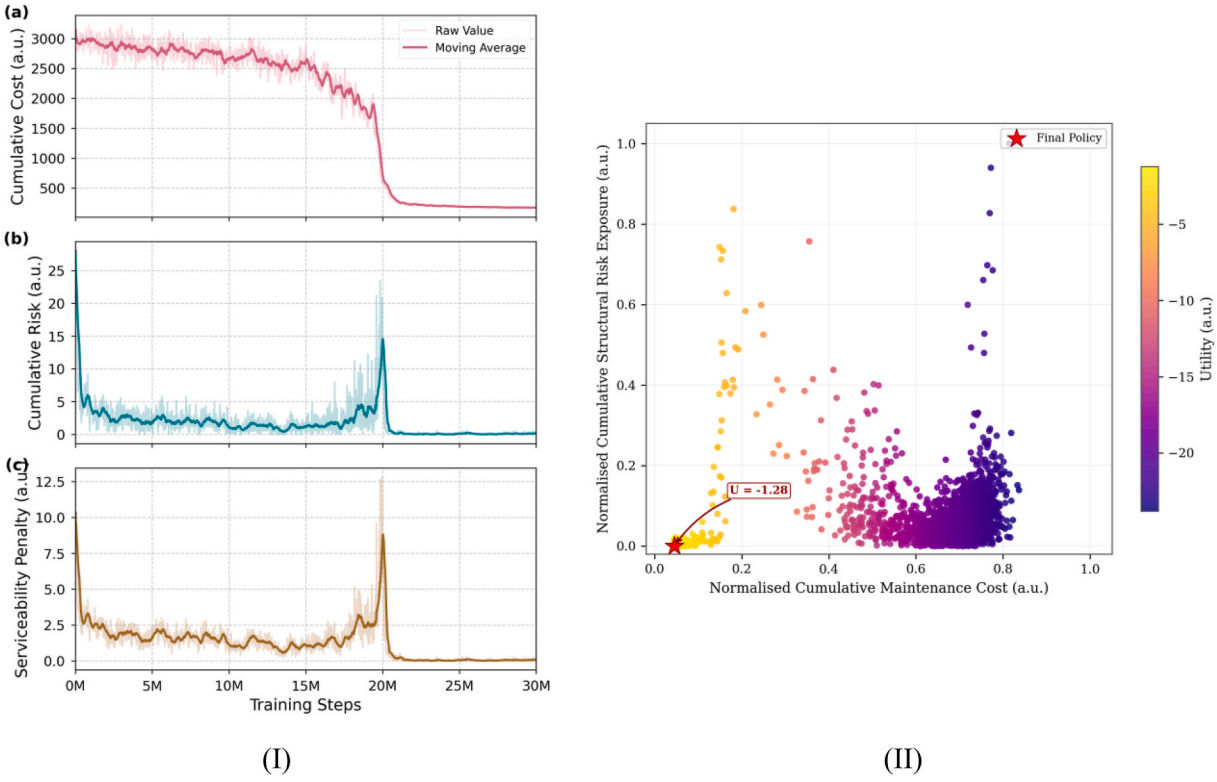


Fig. 11. Training performance of the MO-DCMAC for the McKanes Falls Bridge case study. (I): Evolution of cumulative cost (a), risk (b), and serviceability (c) over 30 million training steps; (II): The policy training trajectory showcasing the training evolution of normalized cost and risk.

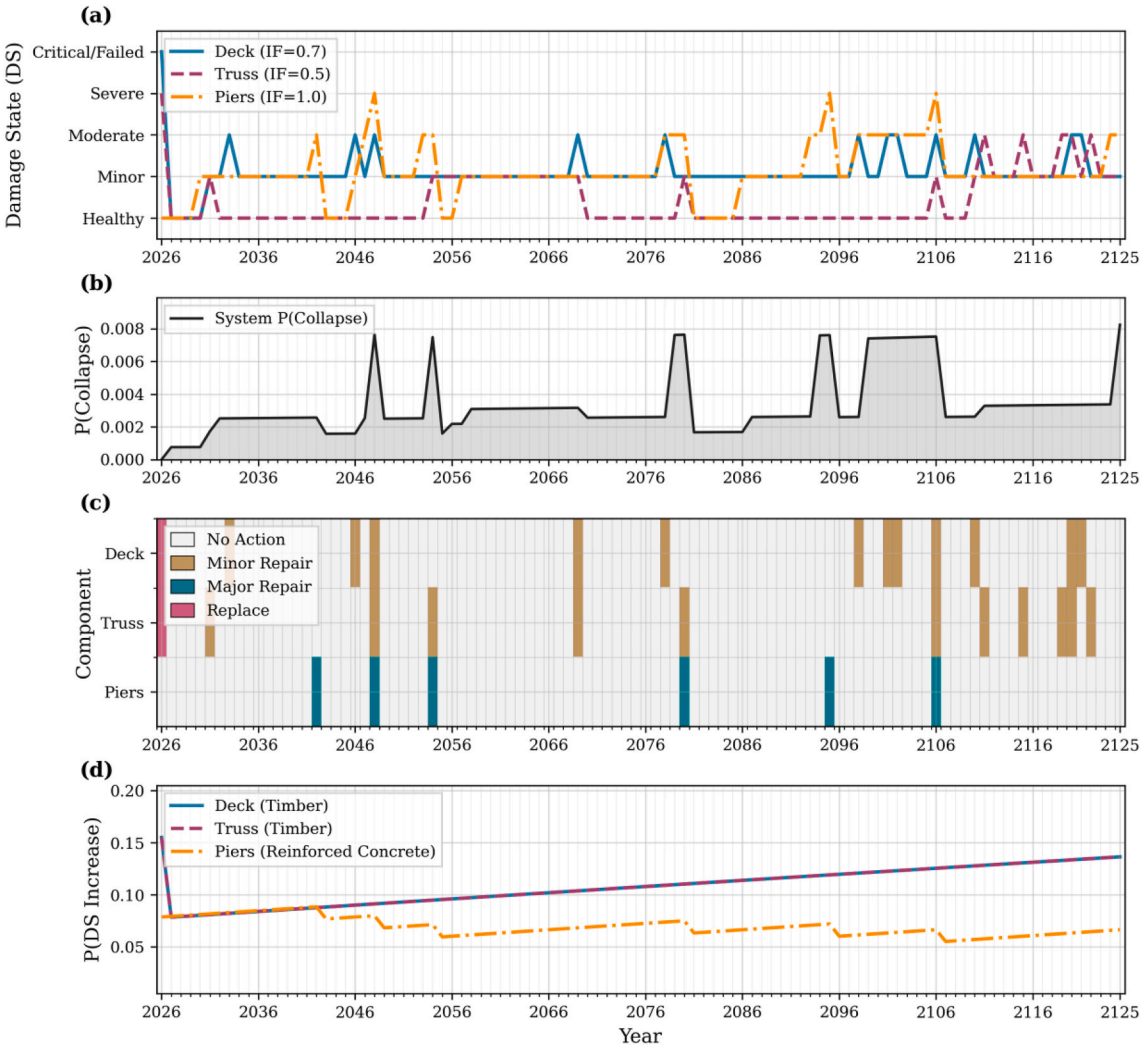


Fig. 12. The results of 100-year maintenance planning for McKanes Falls bridge; (a): Distribution of component damage states; (b): System-level probability of collapse; (c): Distribution of maintenance actions by components; (d): The probability of future deterioration.

agent relies heavily on batched maintenance to minimize mobilization costs, often repairing the deck, truss, and piers in the same year. Furthermore, the agent adapts its repair type to the material properties. The reinforced concrete piers, which possess the highest structural importance (IF = 1.0), are treated exclusively with major repairs to proactively manage their structural age. In contrast, the fast-deteriorating timber components are managed through frequent minor repairs, keeping them fluctuating safely between healthy and moderate damage states without triggering overkill penalties.

The success of this aggressive initial management is reflected in the system's probability of collapse, shown in Fig. 12(b). After the immediate replacements in the first year, the system risk drops dramatically and remains securely constrained below 0.01 for the rest of the century, well beneath the predefined tolerable risk threshold.

Finally, the underlying deterioration probabilities in Fig. 12(d) visually confirm the logic behind the initial replacement strategy. The timber components begin with extremely high inherent probabilities of deterioration due to their advanced age. The immediate, dramatic drop in these curves at the start of the simulation proves that the agent's replacement action effectively eliminated the fundamental drivers of physical decay, paving the way for a century of safe, low-risk operation.

7. Policy comparison

Following the evaluation of the proposed MO-DCMAC framework on the case studies, the same trained RL agent for each scenario was utilized to compare policies against the established rule-based approaches.

7.1. Baseline maintenance policies

To properly evaluate the performance of the proposed MO-DCMAC agent, its decision-making capabilities are benchmarked against several common rule-based policies. These baseline policies reflect established maintenance strategies, ranging from purely passive to proactive, and are frequently used as benchmarks in academic research and asset management guidelines (Ahmad, 2011; Strauss et al., 2023; van Remmerden et al., 2025). This comparative analysis serves to highlight the unique advantages and learned efficiencies of the adaptive, utility-driven RL agent. The policies implemented for this comparison are detailed below:

- **Reactive (Run-to-Failure) Policy**: Under this policy, no maintenance is performed during a component's operational life. An intervention, specifically a replacement, is only carried out once a component has deteriorated to its critical damage state (DS = 4). This

strategy serves as a baseline representing minimal initial investment with maximum risk.

- **Time-Based Policy**: Maintenance actions are performed according to a predetermined, fixed schedule, irrespective of the component's actual physical condition. In this implementation, a minor repair is applied to all components at 10-year intervals, while a major repair is conducted at 30-year intervals. A safety override is included, whereby any component reaching a critical failure state is immediately replaced.
- **Multi-Threshold Condition-Based Policy**: This policy is structured as a dynamic, multi-threshold approach where maintenance decisions are directly triggered by the assessed condition of a component. Specific interventions are matched to predefined DS thresholds. A minor repair is initiated when moderate damage is observed ($DS = 2$), and a major repair is performed for severe damage ($DS = 3$). If a component's condition degrades to a critical state ($DS \geq 4$), it is replaced.
- **Hybrid Policy** (Time-Based Inspection, Condition-Based Action): This policy integrates both time-based and condition-based elements. Component conditions are assessed at fixed 5-year intervals. Following each scheduled inspection, maintenance actions are determined using the same rules as the Multi-Threshold Condition-Based Policy. In the years between these scheduled assessments, no proactive maintenance is performed, though a critical failure override remains active.

7.2. Policy comparison results

The evaluation is based on the averaged results from the Werrington, Mt Emu Creek, and McKanes Falls bridges to comprehensively assess each policy's effectiveness over a 100-year lifecycle. The results of the policy comparison are summarized in Table 8 and illustrated in Fig. 13.

Table 8 reports the mean and standard deviation of the utility score and objectives alongside the average number of interventions over 1000 simulated lifecycles per policy. The RL agent achieves the highest overall utility score (−1.18) among all evaluated strategies. It is followed by the Hybrid, Condition-Based, Time-Based, and Reactive policies in descending order of performance.

The RL agent's superior performance highlights its ability to balance strict safety requirements with realistic economic constraints. While the Condition-Based policy also maintains perfect safety and serviceability levels with zero penalties, it does so at a much higher financial cost. Because the simulation includes fixed mobilization costs for any site visit, the Condition-Based approach is heavily penalized for triggering frequent, isolated repairs at every minor damage threshold.

In contrast, the RL agent learns an optimized, preventative maintenance strategy. It strategically batches maintenance actions by allowing minor damage to accumulate slightly, then dispatching crews to repair multiple components simultaneously. This intelligent clustering reduces the total number of interventions, effectively minimizing expensive setup overheads while keeping the structure strictly within the safe, zero-penalty zone.

The remaining baseline strategies demonstrate distinct operational flaws. The Reactive policy yields the lowest utility due to catastrophic penalties from unmitigated structural risk and severe serviceability failures. The Time-Based approach performs better but still incurs substantial penalties because its rigid schedule ignores the actual physical condition of the aging components. While the Hybrid policy offers a closer approximation of the optimal trade-off, it lacks the dynamic adaptability of the RL agent to truly minimize long-term expenditures.

Fig. 13(a) illustrates the evolution of the average system DS over time for each policy. The RL agent maintains the bridge at a consistently lower damage index compared to all benchmark policies. By keeping the structure in this healthy zone, the agent avoids the severe financial penalties associated with critical deterioration. In contrast, the Time-Based profile exhibits sawtooth patterns corresponding to scheduled repairs, and the Reactive approach allows unsafe levels of damage to accumulate before intervening.

The composition of interventions is shown in Fig. 13(b). The Condition-Based policy triggers an excessive number of minor repairs because it strictly reacts to every small defect. The RL agent learns a more sophisticated strategy. It employs a balanced mix of minor patches and impactful major repairs, recognizing that occasional structural resets are more economical than constant minor interventions that repeatedly trigger mobilization setup fees.

Finally, Fig. 13(c) highlights the relationship between total interventions and overall effectiveness. The RL agent is positioned closest to the ideal corner of the trade-off space. It achieves the highest mean utility by executing fewer total interventions than the Condition-Based approach, which wastes resources on excessive setup costs. At the same time, the RL agent remains proactive enough to completely prevent the catastrophic failure penalties that ruin the Reactive strategy. This demonstrates that the RL framework successfully learns a highly optimized, context-aware maintenance schedule that rigid rule-based systems cannot replicate.

8. Parameter analysis

The MO-DCMAC framework requires the specification of several principal parameters. These parameters directly influence the trade-off structure embedded in the optimization process and, consequently, the characteristics of the learned maintenance policy. To address this issue, some sensitivity analysis tests are conducted in this section to evaluate how variations in key parameters affect the performance of the proposed framework.

To understand how policy evaluation changes in response to variations in utility weights and penalty thresholds, heterogeneous weight configurations are assigned to the objective functions to represent distinct stakeholder preferences. Each configuration can be interpreted as a specific decision-making profile. For instance, a risk-averse stakeholder assigns greater weight to reliability and risk mitigation, whereas a risk-seeking stakeholder prioritizes cost efficiency and operational flexibility. These preference structures are summarized in Table 9.

To evaluate the robustness of the proposed framework, the utility weighting schemes were systematically varied to represent different stakeholder priorities. This preference analysis was conducted on the Werrington bridge case study using 100 evaluation episodes. Fig. 14 presents the comparative performance of all policies under these diverse decision-making profiles.

As shown in the bar chart and heatmap in Fig. 14(a) and (b), the RL agent consistently achieves the highest mean utility across every

Table 8
Mean and standard deviation of utility, objectives, and intervention counts for each policy over 1000 simulated lifecycles.

Policy	Utility	Cost	Risk	Serviceability	Intervention
RL Agent	-1.18 ± 0.18	16207 ± 2490	0.00	0.00	26.80
Reactive	-7.38 ± 3.41	15980 ± 2790	-40748 ± 24476	-17683 ± 6233	6.17
Time-Based	-2.45 ± 1.08	20487 ± 2324	-5829 ± 7529	-3551 ± 2453	31.47
Condition-Based	-1.78 ± 0.25	24259 ± 3480	0.00	0.00	34.74
Hybrid	-1.77 ± 0.48	17009 ± 2105	-3208 ± 2978	-1906 ± 998	23.59

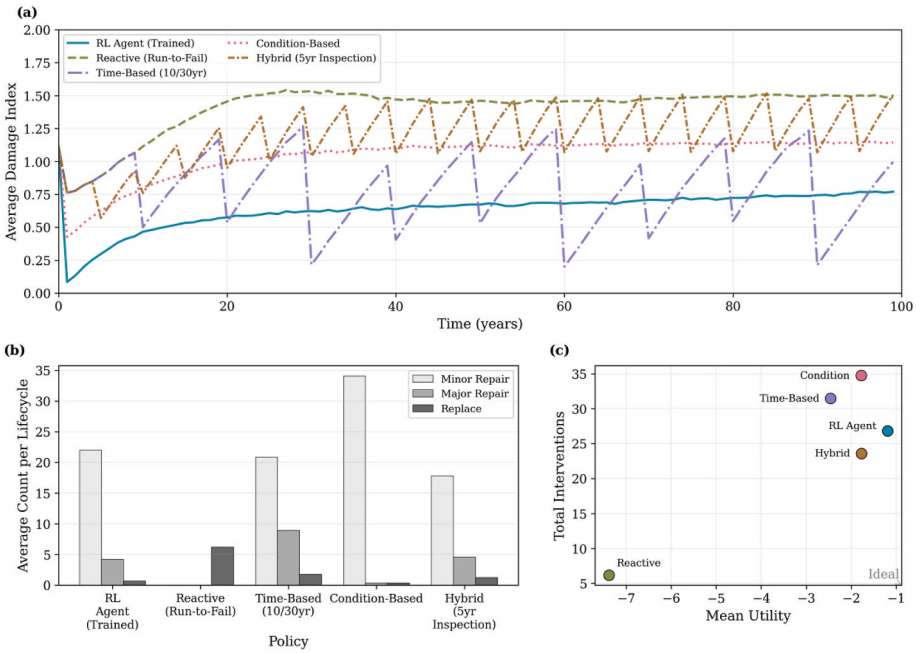


Fig. 13. The results comparison of maintenance policies; (a): Average system damage index over time (Lower is better); (b): Average count of intervention types per lifecycle; and (c): The trade-off between total interventions (Lower is better) and mean utility (Higher is better).

Table 9
Stakeholder preference profiles and corresponding utility weight configurations used in the sensitivity analysis.

Stakeholder Profile	Cost Weight	Risk Weight	Serviceability Weight
Base (Training)	1.10	1.00	0.95
Balanced	1.00	1.00	1.00
Cost-Averse	1.50	1.00	1.00
Very Cost-Averse	2.00	0.80	0.80
Safety-First	1.00	1.50	1.00
Risk-Seeking	1.20	0.60	1.00
Service-Focused	1.00	1.00	1.50
Conservative	1.30	1.30	0.80

stakeholder configuration. Whether a profile strictly prioritizes safety, minimizes cost, or focuses on serviceability, the RL policy remains the top performing strategy. This stability confirms that the agent successfully internalized the complex relationship between all three competing objectives rather than overfitting to a single goal.

Furthermore, Fig. 14(c) illustrates the balance between mean cumulative cost and risk penalties. The RL agent occupies the most favourable position in this space, aligning closely with the ideal optimal direction. While the reactive strategy suffers massive risk penalties and the condition-based approach incurs excessive financial costs, the RL framework provides a highly efficient and balanced maintenance plan.

Overall, these results shows that variations in utility weights and stakeholder priorities shift the quantitative utility values, yet do not alter the qualitative dominance of the RL-based policy. This confirms the robustness, stability, and generalizability of the proposed framework in accommodating diverse decision-making preferences while maintaining superior performance within the evaluated trade-off space.

A second sensitivity analysis is conducted to investigate the impact of the tolerable risk threshold parameter on policy performance. In the simulation environment, massive safety penalties are only triggered if the overall system collapse probability exceeds this specific limit. This analysis reveals whether a maintenance strategy is genuinely safe or if it relies on relaxed regulations to avoid penalties.

Fig. 15 presents the comparative results for different risk threshold values in the Werrington bridge case study. Fig. 15(a) shows that as the acceptable risk limit increases, the mean utility naturally improves for all policies because the safety constraints become less punishing. However, the RL agent consistently outperforms the reactive, time-based, condition-based, and hybrid approaches across the entire range of thresholds. The reactive strategy performs particularly poorly under strict thresholds due to substantial risk penalties, whereas the rule-based preventive strategies demonstrate moderate improvements but do not surpass the RL-based policy.

Fig. 15(b) illustrates the corresponding mean risk penalties. As expected, higher thresholds reduce the imposed penalties for all policies. Nevertheless, the RL agent maintains significantly lower risk penalties compared to the reactive and time-based policies, particularly under stringent safety constraints. This indicates that the RL-based strategy effectively anticipates deterioration and schedules interventions in a manner that controls risk proactively rather than reactively.

Overall, these results confirms that the proposed framework is robust not only to variations in stakeholder preference weights but also to changes in regulatory or safety tolerance parameters. The RL agent consistently achieves superior utility performance while maintaining controlled risk levels across diverse threshold settings.

9. Potential limitations

Although the suggested RL-DSS framework offers adaptive, informed and cost-effective decision-making process and serves as a robust foundation for the development of intelligent, data-driven maintenance systems, specific limitations necessitate consideration to guide prospective future improvements.

The practical deployment of the RL-DSS in the management of critical bridge infrastructures remains challenging due to regulatory constraints, limited interpretability and reduced stakeholders trust in automated policy decisions. Automated policies may be difficult to fully explain or justify, potentially reducing trust among engineers and decision-makers. Consequently, implementation of RL-DSS framework across large-scale, critical, and iconic bridges within DTs platform and the evaluation of its performance in comparison to the stablished policies remains constrained.

Moreover, despite recent research advancements, real-time and

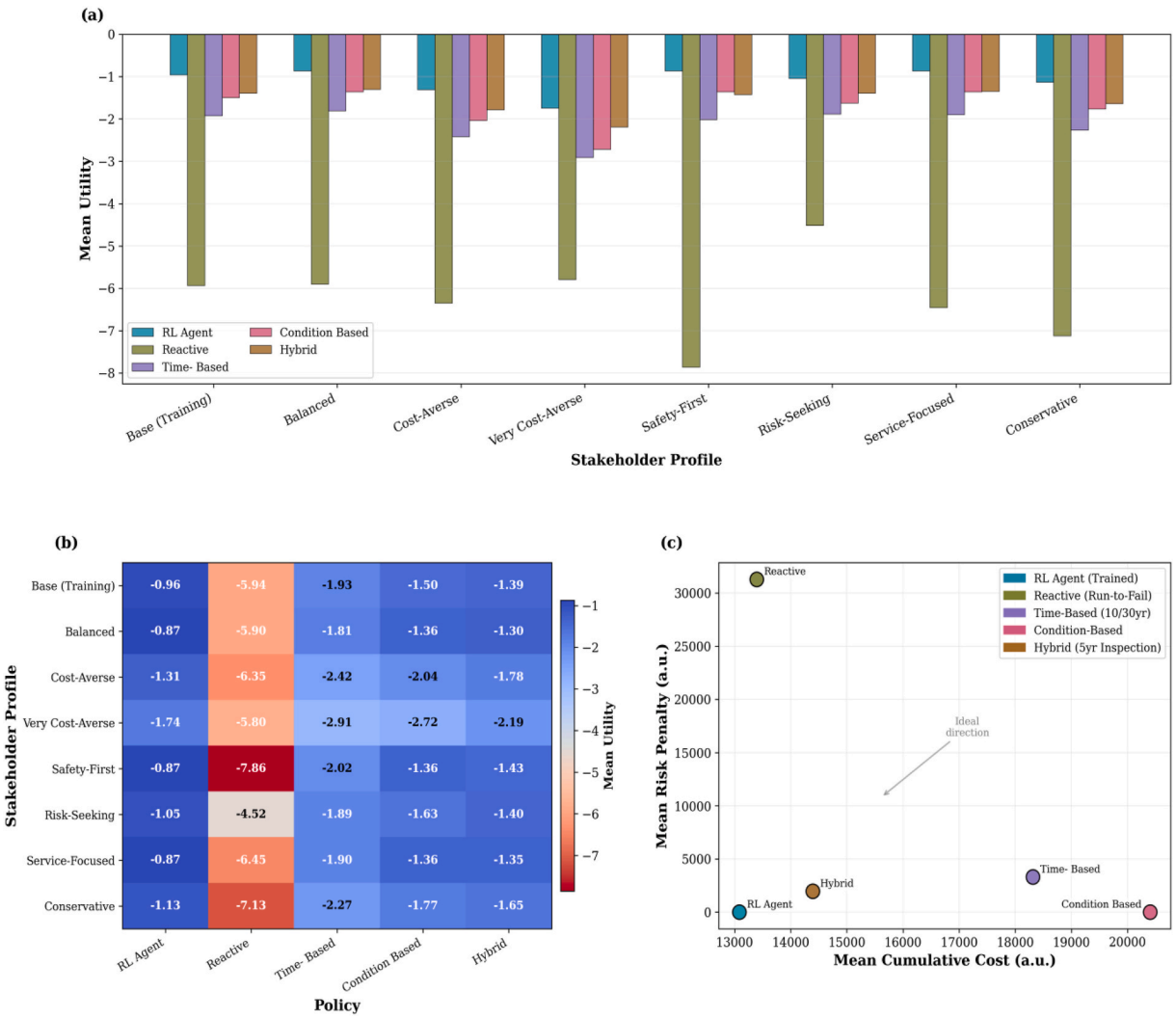


Fig. 14. Performance comparison of maintenance policies across different stakeholder profiles for the Werrington bridge. (a) Mean utility, (b) utility score heatmap, and (c) trade-off between mean cumulative cost and risk.

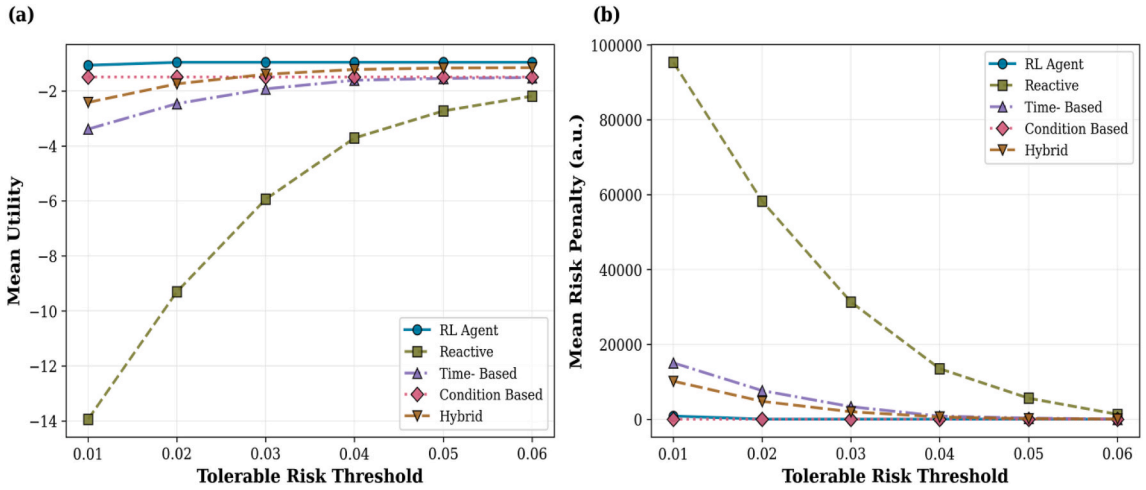


Fig. 15. Performance of maintenance policies under varying tolerable risk thresholds for the Werrington bridge. (a) Mean utility scores and (b) mean risk penalties.

accurate modelling of structural responses to evaluate bridge condition information within DT platforms, which is an essential input for decision-making within the proposed RL-DSS framework, remains a challenging task, especially under complex loading conditions or unforeseen events. Inaccuracies in these inputs may result in suboptimal policy recommendations. In addition, the RL-DSS may generate

recommendations that, while theoretically optimal, might be impractical or unsafe for real-world implementation if not appropriately constrained.

Despite the promising results, the simulated environment applied in this study represents a simplification of reality, and its parameters could be further calibrated to enhance the real-world applicability of the learned policy. The current framework does not account for complex logistical constraints such as annual network level budget limits, contractor availability, or the indirect costs of traffic disruptions. Incorporating these broader economic and logistical factors would require a more expansive simulation environment.

Finally, the performance of the framework is contingent on the calibration of the underlying deterioration and risk models, which require comprehensive real-world data for maximum fidelity. Consequently, while the RL agent converged on logical and optimal policies within the defined simulation, the direct applicability of these specific policies to real-world operations would require further refinement of the environmental parameters.

10. Conclusions

This research introduces an advanced Reinforcement Learning-based Decision Support System (RL-DSS) framework designed for bridge Digital Twins (DTs) to support optimized maintenance planning for sustainable infrastructure with multiple interacting components. This innovative framework addresses the need for robust management systems under deterioration, increasing complexity, and constrained financial resources. The proposed framework leverages a Multi-Objective Deep Centralized Multi-Agent Actor-Critic (MO-DCMAC) algorithm to simultaneously balance the competing objectives of maintenance cost, structural failure risk, and structural serviceability. By formulating the problem as a Multi-Objective Markov Decision Process (MOMDP) with multiple criteria, the system learns adaptive policies within a simulated environment, demonstrating a pathway for application within bridge DTs.

This research provides an implementation-level extension of MO-DCMAC to a three-objective bridge maintenance problem involving cost, risk, and serviceability. Under the fixed linear utility function adopted in this study, the actor objective is equivalent to conventional scalarized expected-return reinforcement learning. The primary contribution lies in retaining separate objective-return dimensions within the distributional critic and developing a computationally tractable three-objective Bellman projection. This enables independent monitoring of cost, risk, and serviceability, while maintaining a single scalar utility for policy optimization.

The results of three distinct real-world bridge case studies demonstrated the framework's adaptability and effectiveness across various structural types, ages, and environmental conditions. Comparative analysis demonstrated the clear superiority of the RL agent over traditional rule-based benchmark policies. Unlike reactive, rigid time-based, or strict condition-based strategies, the intelligent agent learned a highly proactive approach. It strategically clustered maintenance actions to minimize expensive setup costs while consistently preventing components from reaching critical and heavily penalized deterioration states.

The proposed RL-DSS establishes a fundamental and extensible approach with significant potential for future advancements in smart infrastructure management. Future research should prioritize improving both the realism and practical applicability of the framework by expanding the simulation environment to account for broader logistical constraints. This includes incorporating annual network level budget limits, contractor availability, and the indirect costs associated with traffic disruptions. In addition, extending the framework to explicitly address partial observability and discrete inspection intervals through a POMDP formulation or belief-state construction would further enhance its practical applicability under real-world inspection uncertainty. Calibrating the model with historical inspection data and real-time monitoring information would significantly improve the accuracy and reliability of the learned policies. The framework could also be extended to accommodate additional objectives, enabling a more comprehensive approach to bridge maintenance optimization.

CRediT authorship contribution statement

Vahid Mousavi: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Validation, Visualization, Writing – original draft, Writing – review & editing. **Maria Rashidi:** Conceptualization, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Supervision, Validation, Writing – review & editing. **Shayan Ghazimoghadam:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Validation, Writing – original draft, Writing – review & editing. **Masoud Mohammadi:** Conceptualization, Formal analysis, Methodology, Supervision, Validation, Writing – review & editing. **Bijan Samali:** Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors express their deep appreciation for the support provided by Australian Research Council through Industry Transformation Research Hub for Resilient and Intelligent Infrastructure Systems (RIIS) and the IPWEA Road and Transport Directorate of NSW.

Data availability

Data will be made available on request.

References

Ahmad, A.S., 2011. Bridge Preservation Guide: Maintaining a State of Good Repair Using Cost Effective Investment Strategies. Federal Highway Administration, United States.
Andersen, N.H., 1990. DANBRO—A bridge management system for many levels. Bridge Evaluation, Repair and Rehabilitation. Springer, pp. 11–21.
Australian Standard, A., 2004. 5100: Bridge Design. Standards Australia International (SAI), Sydney.
Brighenti, F., Caspani, V.F., Costa, G., Giordano, P.F., Limongelli, M.P., Zonta, D., 2024. Bridge management systems: a review on current practice in a digitizing world. Eng. Struct. 321, 118971.
Chase, S.B., Adu-Gyamfi, Y., Aktan, A., Minaie, E., 2016. Synthesis of National and International Methodologies Used for Bridge Health Indices.
de Jonge, B., Teunter, R., Tinga, T., 2017. The influence of practical factors on the benefits of condition-based maintenance over time-based maintenance. Reliab. Eng. Syst. Saf. 158, 21–30.
Du, A., Ghavidel, A., 2022. Parameterized deep reinforcement learning-enabled maintenance decision-support and life-cycle risk assessment for highway bridge portfolios. Struct. Saf. 97, 102221.
Ellenberg, A., Kontsos, A., Moon, F., Bartoli, I., 2016. Bridge related damage quantification using unmanned aerial vehicle imagery. Struct. Control Health Monit. 23, 1168–1179.
Frangopol, D.M., Kim, S., 2022. Bridge Safety, Maintenance and Management in a life-cycle Context. CRC Press.
Ghafoori, M., Abdallah, M., Ozbek, M.E., 2024. Machine learning–based bridge maintenance optimization model for maximizing performance within available annual budgets. J. Bridge Eng. 29, 04024011.
Görçün, Ö.F., Özçalıcı, M., Gurler, H.E., Pamucar, D., Simic, V., 2025. Promoting sustainable urban mobility: an integrated fuzzy decision-making model for assessing autonomous bus alternatives. Eng. Appl. Artif. Intell. 156, 111034.
Hawk, H., Small, E.P., 1998. The BRIDGIT bridge management system. Struct. Eng. Int. 8, 309–314.
Hayes, C.F., Rădulescu, R., Bargiacchi, E., Källström, J., Macfarlane, M., Reymond, M., Verstraeten, T., Zintgraf, L.M., Dazeley, R., Heintz, F., 2022a. A practical guide to multi-objective reinforcement learning and planning. Aut. Agents Multi-Agent Syst. 36, 26.

Hayes, C.F., Rădulescu, R., Bargiacchi, E., Källström, J., Macfarlane, M., Reymond, M., Verstraeten, T., Zintgraf, L.M., Dazeley, R., Heintz, F., 2022b. A practical guide to multi-objective reinforcement learning and planning: CF Hayes et al. Aut. Agents Multi-Agent Syst. 36, 26.
He, C., Liu, M., Hsiang, S.M., Pierce, N., Megahed, S., Godfrey, A., 2025. An ontological knowledge-driven smart contract framework for implicit bridge preservation decision making. J. Construct. Eng. Manag. 151, 04025008.
Heng, J., Dong, Y., Lai, L., Zhou, Z., Frangopol, D.M., 2024. Digital twins-boosted intelligent maintenance of ageing bridge hangers exposed to coupled corrosion–fatigue deterioration. Autom. ConStruct. 167, 105697.
Hopper, T., Langlois, A., Murphy, T., 2022. Service Life Design Reference Guide. Federal Highway Administration, Washington DC.
Jaafaru, H., Agbelie, B., 2022. Bridge maintenance planning framework using machine learning, multi-attribute utility theory and evolutionary optimization models. Autom. ConStruct. 141, 104460.
Lei, X., Dong, Y., Frangopol, D.M., 2023. Sustainable life-cycle maintenance policymaking for network-level deteriorating bridges with a convolutional autoencoder–structured reinforcement learning agent. J. Bridge Eng. 28, 04023063.
Lei, X., Dong, Y., Frangopol, D.M., 2025. Integration of inspection and monitoring data for RL-enhanced sustainable life-cycle management of infrastructure networks. Struct. Infrastruct. Eng. 1–15.
Lei, X., Xia, Y., Deng, L., Sun, L., 2022. A deep reinforcement learning framework for life-cycle maintenance planning of regional deteriorating bridges using inspection data. Struct. Multidiscip. Optim. 65, 149.
Lowe, R., Wu, Y.I., Tamar, A., Harb, J., Pieter Abbeel, O., Mordatch, I., 2017. Multi-agent actor-critic for mixed cooperative-competitive environments. Adv. Neural Inf. Process. Syst. 30.
Mohammadi, M., Rashidi, M., Yu, Y., Samali, B., 2023. Integration of TLS-derived bridge information modeling (BrIM) with a Decision support system (DSS) for digital twinning and asset management of bridge infrastructures. Comput. Ind. 147, 103881.
Moslem, S., Tezel, B.T., Kinay, A.O., Pilla, F., 2024. A hybrid approach based on magnitude-based fuzzy analytic hierarchy process for estimating sustainable urban transport solutions. Eng. Appl. Artif. Intell. 137, 109112.
Mousavi, V., Rashidi, M., Ghazimoghadam, S., Mohammadi, M., Samali, B., Devitt, J., 2025a. Transformer-based time-series GAN for data augmentation in bridge digital twins. Autom. ConStruct. 175, 106208.
Mousavi, V., Rashidi, M., Mohammadi, M., Samali, B., 2024. Evolution of digital twin frameworks in bridge management: review and future directions. Remote Sens. 16, 1887.
Mousavi, V., Rashidi, M., Shayan, G., Samali, B., 2025b. A data-driven digital twin model for bridge health monitoring using feature fusion and unsupervised deep learning. Inf. Fusion, 103534.
Rashidi, M., 2013. Decision Support System for Remediation of Concrete Bridges. University of Wollongong.
Rashidi, M., Samali, B., Sharafi, P., 2016a. A new model for bridge management: part A: condition assessment and priority ranking of bridges. Aust. J. Civ. Eng. 14, 35–45.
Rashidi, M., Samali, B., Sharafi, P., 2016b. A new model for bridge management: part B: decision support system for remediation planning. Aust. J. Civ. Eng. 14, 46–53.
Rashidi, M., Gibson, P., 2011. Proposal of a Methodology for Bridge Condition Assessment.
Rashidi, M., Gibson, P., 2012. A Methodology for Bridge Condition Evaluation.
Saha, A., Görçün, Ö.F., Pamucar, D., Arya, L., Simic, V., 2024. Evaluation of shared micro-mobility systems for sustainable cities by using a consensus-based fermatean fuzzy multiple objective optimization and full multiplicative form. Eng. Appl. Artif. Intell. 134, 108662.
Saifullah, M., Papakonstantinou, K., Andriotis, C., Stoffels, S., 2024. Multi-agent deep reinforcement learning with centralized training and decentralized execution for transportation infrastructure management. arXiv preprint arXiv:2401.12455.
Shahrivar, F., Sidiq, A., Mahmoodian, M., Jayasinghe, S., Sun, Z., Setunge, S., 2025. AI-based bridge maintenance management: a comprehensive review. Artif. Intell. Rev. 58, 135.
Sood, S.K., 2024. A scientometric analysis of quantum driven innovations in intelligent transportation systems. Eng. Appl. Artif. Intell. 138, 109258.
Strauss, A., Bergmeister, K., Vliet, A.B.v., Daró, P., Zimmermann, T., van Meerveld, H., 2023. Strategies for low limit maintenance thresholds and condition states for bridge structures. ce/papers 6, 578–586.
Sutton, R.S., Barto, A.G., 1998. Reinforcement Learning: an Introduction. MIT press Cambridge.
Tang, H., Wang, J., Wu, J., Zhao, Y., Chen, J., Liang, F., Zhang, Z., 2025. The nexus of intelligent transportation: a lightweight Bi-input fusion detection model for autonomous-rail rapid transit. Eng. Appl. Artif. Intell. 139, 109705.
Thompson, P.D., Small, E.P., Johnson, M., Marshall, A.R., 1998. The pontis bridge management system. Struct. Eng. Int. 8, 303–308.
Valenzuela, S., de Solminihac, H., Echaveguren, T., 2010. Proposal of an integrated index for prioritization of bridge maintenance. J. Bridge Eng. 15, 337–343.
van Remmerden, J., Kenter, M., Roijers, D.M., Andriotis, C., Zhang, Y., Bukhsh, Z., 2025. Deep multi-objective reinforcement learning for utility-based infrastructural maintenance optimization. Neural Comput. Appl. 1–24.
Wang, Y., Wang, Y., Ni, J., Zhang, H., 2024. Reliability-centric maintenance planning for bridge infrastructure: a novel method based on improved electric fish optimization. Buildings 14, 3583.
Watkins, C.J., Dayan, P., 1992. Technical note Q-learning. Mach. Learn.
Wei, S., Bao, Y., Li, H., 2020. Optimal policy for structure maintenance: a deep reinforcement learning framework. Struct. Saf. 83, 101906.
Wu, C., Wu, P., Wang, J., Jiang, R., Chen, M., Wang, X., 2021. Critical review of data-driven decision-making in bridge operation and maintenance. Struct. Infrastruct. Eng. 18, 47–70.
Xin, G., Liang, Z., Hu, Y., Long, G., Zhang, Y., Liang, P., 2023. Modeling the optimal maintenance strategy for bridge elements based on agent sequential decision making. Appl. Sci. 14, 14.
Xu, G., Guo, F., 2022. Sustainability-oriented maintenance management of highway bridge networks based on Q-learning. Sustain. Cities Soc. 81, 103855.
Zonta, T., Da Costa, C.A., da Rosa Righi, R., de Lima, M.J., Da Trindade, E.S., Li, G.P., 2020. Predictive maintenance in the industry 4.0: a systematic literature review. Comput. Ind. Eng. 150, 106889.

Glossary

AADT: Average Annual Daily Traffic
BMS: Bridge Management Systems
BrIM: Bridge Information Modelling
CF: Causal Factors
DCMAC: Deep Centralized Multi-Agent Actor-Critic
DL: Deep Learning
DQN: Deep Q-Network
DTs: Digital Twins
DS: Damage State
DSF: Damage-Sensitive Features
DSS: Decision Support System
DRL: Deep Reinforcement Learning
ESR: Expected Scalarized Return
KL: Kullback-Leibler
IF: Importance Factor
MA: Multi-Agent
MDP: Markov Decision Process
MO-DCMAC: Multi-Objective Deep Centralized Multi-Agent Actor-Critic
MODRL: Multi-Objective Deep Reinforcement Learning
MOMDP: Multi-Objective Markov Decision Process
MORL: Multi-Objective Reinforcement Learning
MV: Material Vulnerability
POMDP: Partially Observable Markov Decision Process
RL: Reinforcement learning
RL-DSS: Reinforcement Learning-based Decision Support System